

EE104 Fundamentals of Electric Circuits (Lesson 10)

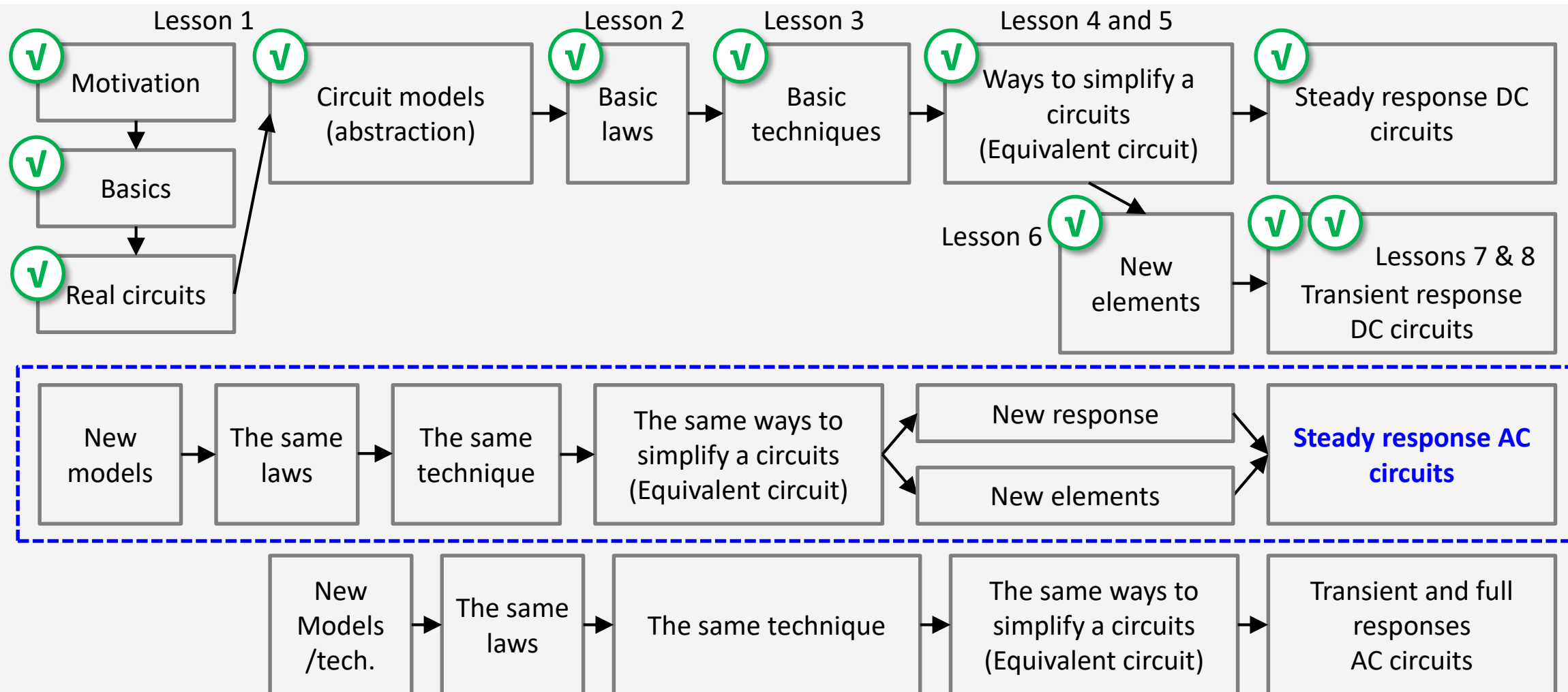
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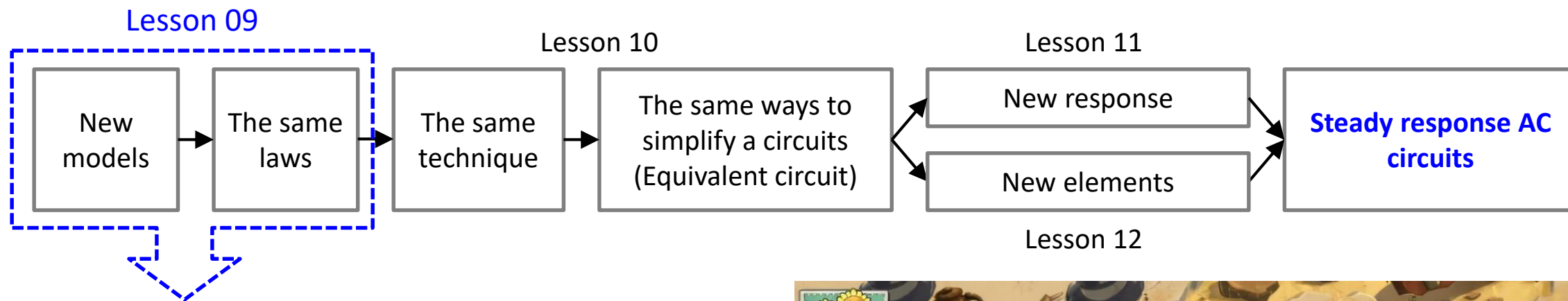
Review of Lessons 1-8 (b)

- EE104 road map



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- EE104 road map



- Sinusoids and phasors
- AC circuits in frequency domain
- Basic laws
- Phase shifters



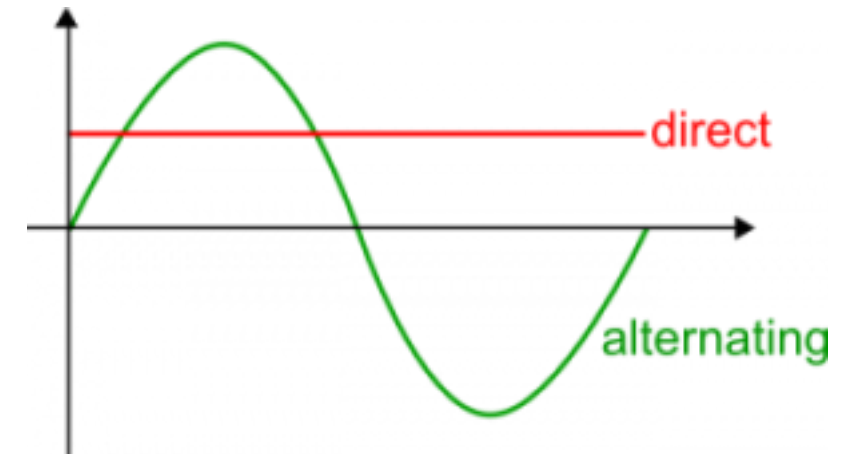
Lesson 9 Sinusoids and Phasors

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9.4	Impedances and admittances in AC	Slides 43-54
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9.1 Introduction (a)

- Sinusoids and AC circuits

- a) In lessons 1 to 8, our analysis has been limited for the most part to dc circuits, which are excited by constant or time-invariant sources, for both pedagogic and historical reasons.
- b) A sinusoid (正弦信号) is a signal that has the form of the sine or cosine function.
- c) A sinusoidal current is usually referred to as alternating current (ac). Such a current reverses at regular time intervals and has alternately positive and negative values. 交流电指周期性正弦型电流; 电流方向交替变化, 取值有正有负。
- d) In contrast, the direct current (dc) remains constant all the time. In other words, the charges flow always in the same direction at the same rate. 直流电流的方向保持不变。
- e) Circuits driven by sinusoidal current or voltage sources are called ac circuits.

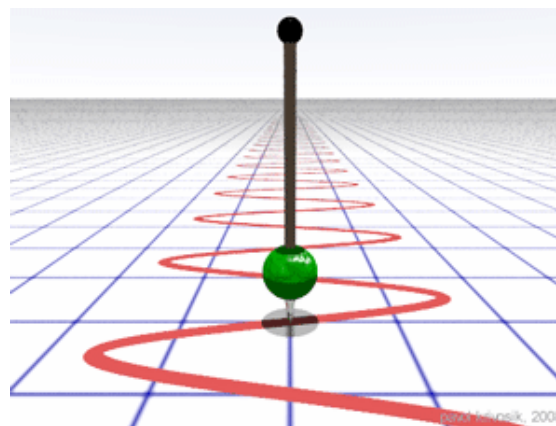


9.1 Introduction (b)

- Why AC ?

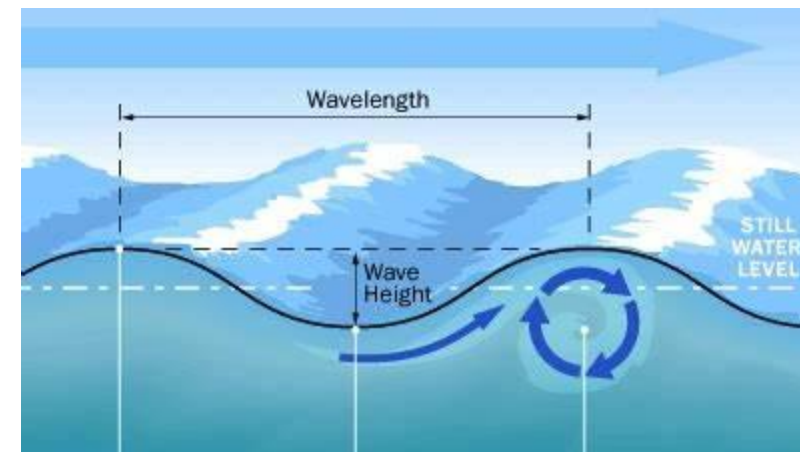
(a)

Many signals are naturally sinusoidal.



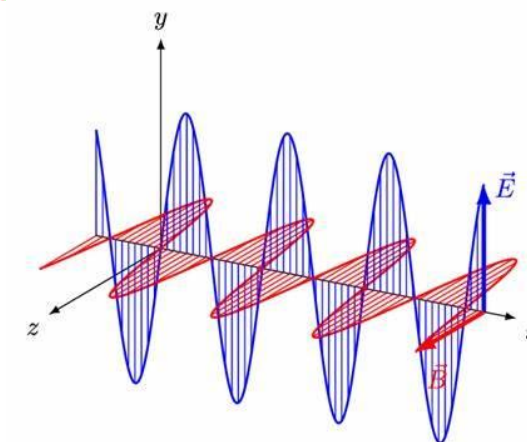
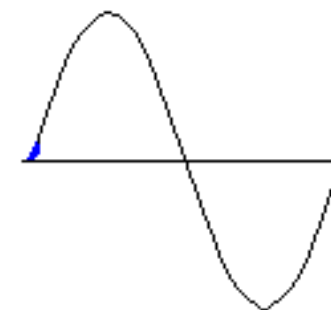
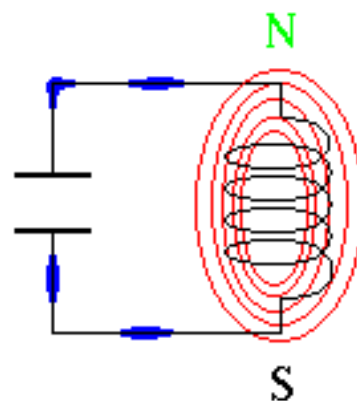
(b)

Sinusoidal signals are easy to generate and transmit.



(c)

Any practical periodic signals can be synthesized by adding up a number of sinusoids of different frequencies.



9.1 Introduction (c)

- Why AC ?

(a)

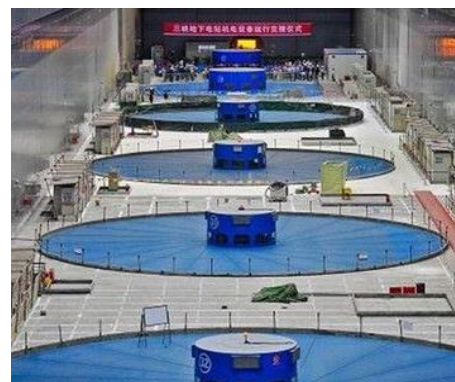
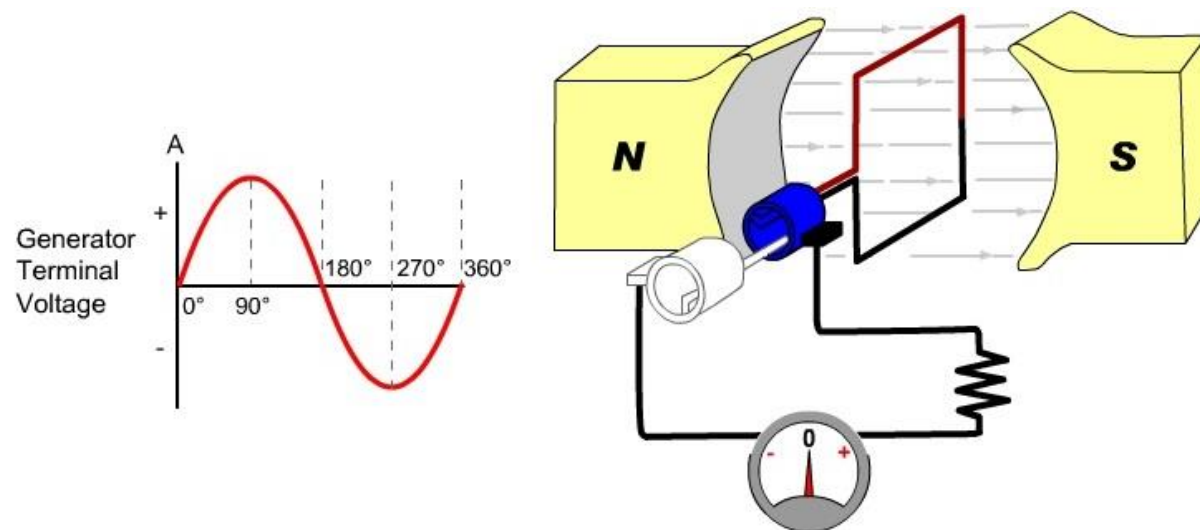
Many signals are naturally sinusoidal.

(b)

Sinusoidal signals are easy to generate and transmit.

(c)

Any practical periodic signals can be synthesized by adding up a number of sinusoids of different frequencies.



9.1 Introduction (d)

- Why AC ?

(a)

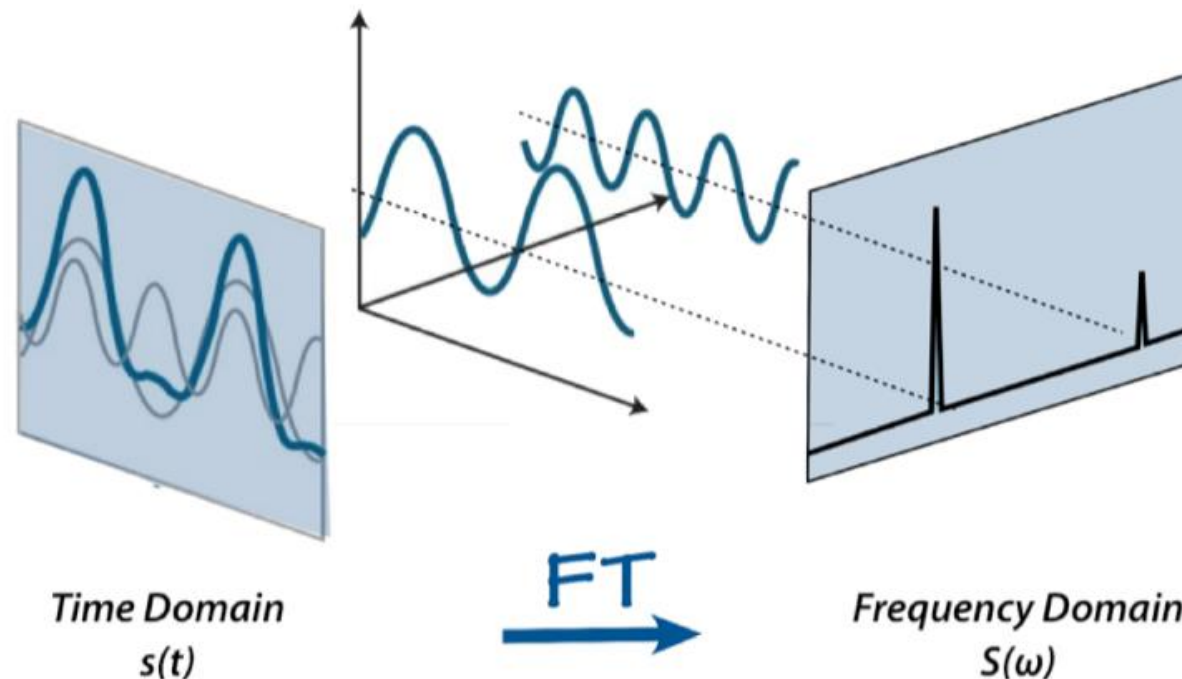
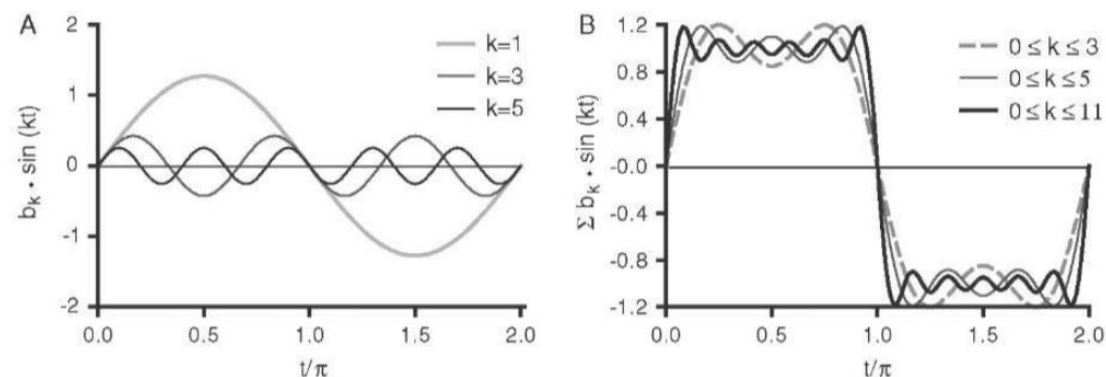
Many signals are naturally sinusoidal.

(b)

Sinusoidal signals are easy to generate and transmit.

(c)

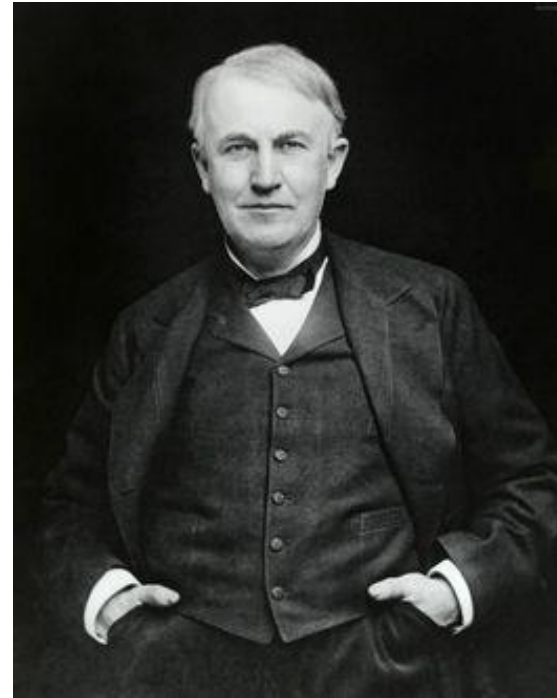
Any practical periodic signals can be synthesized by adding up a number of sinusoids of different frequencies.



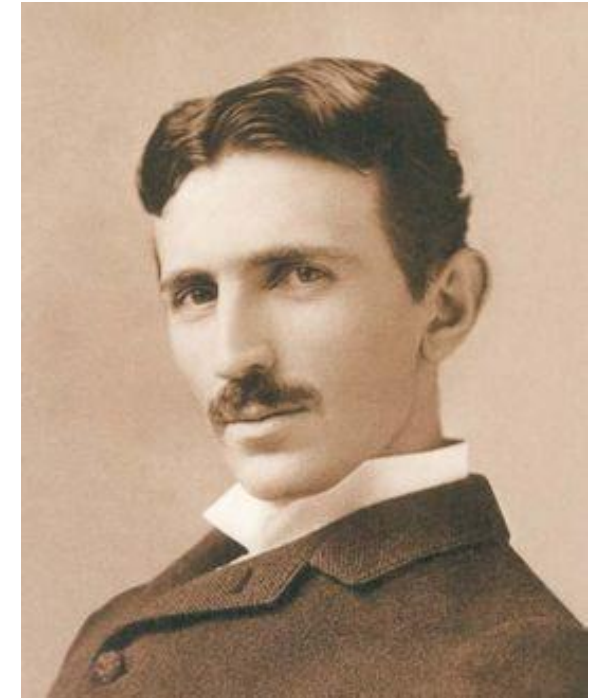
9.1 Introduction (e)

- AC versus DC

- a) In the late 1800's there was a battle between proponents of DC and AC. While Thomas Edison supports DC, Nikola Tesla supports AC.
- b) At that time, one key issue is how to increase the voltage for dc.
- c) The success of induction motor and polyphase generation and distribution systems lead to the domination of ac as the prime energy sources.



Thomas Alva Edison



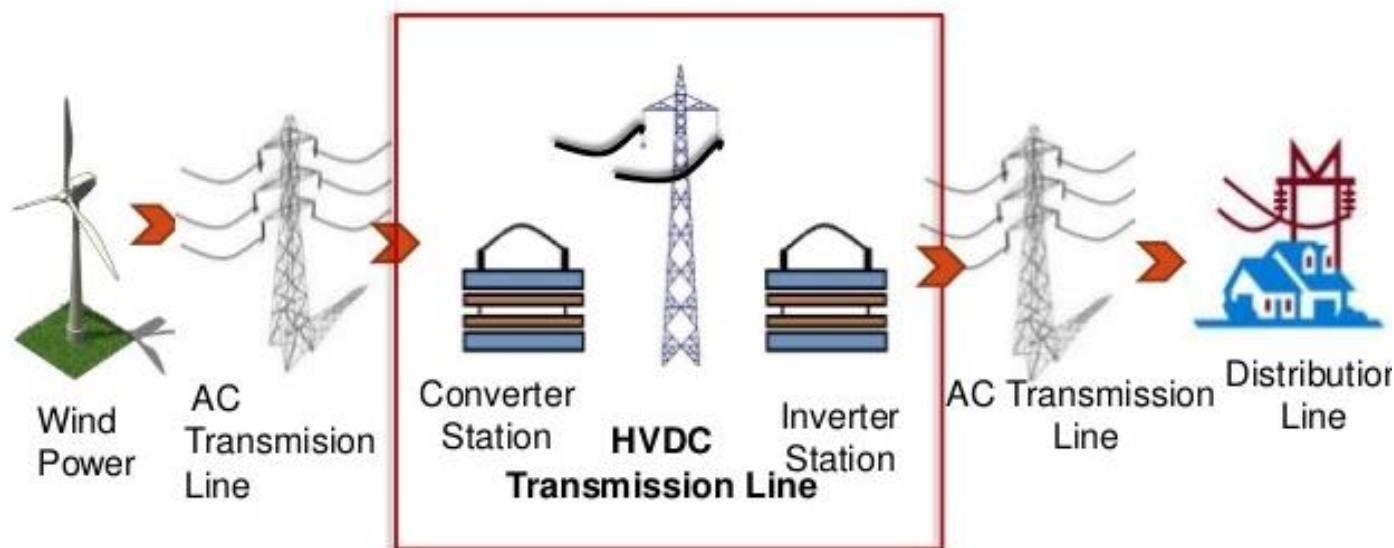
Nikola Tesla

9.1 Introduction (f)

- DC is coming back in many applications

For examples:

- a) High-voltage DC: 高压直流输电系统
- b) Medium-voltage DC: 中压直流综合电力系统
- c) Medium/low-voltage DC: 电动车直流快充

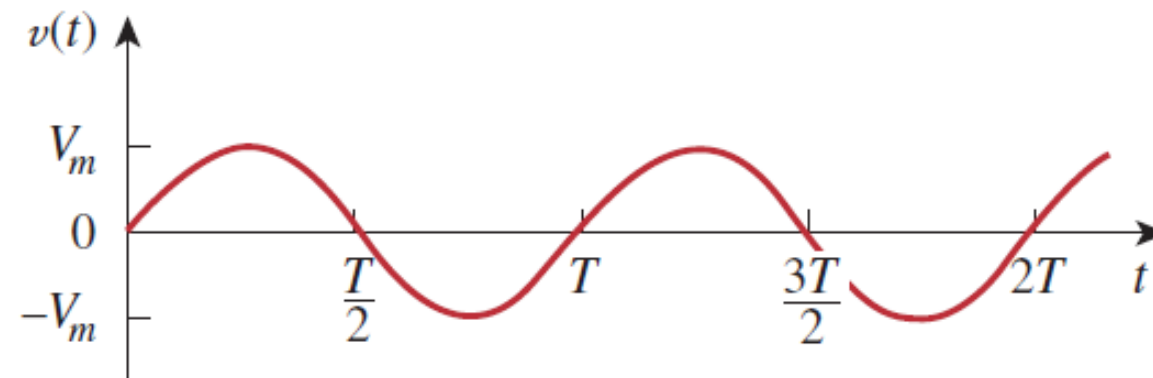
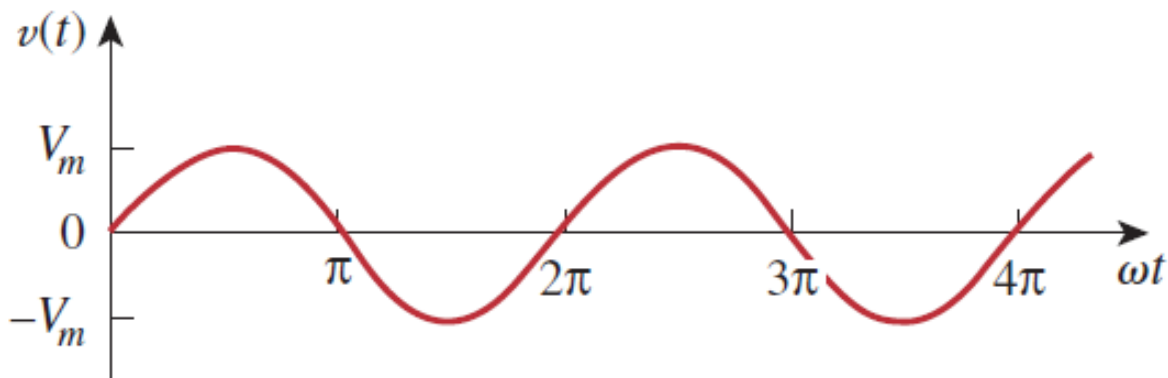


Lesson 9 Sinusoids and Phasors

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9.2 Sinusoids (a)

- Sinusoids (正弦量)



$$v(t) = V_m \sin(\omega t + \phi)$$

Period (周期): $T = \frac{2\pi}{\omega}$

- a) V_m : the amplitude (振幅), unit: V
 - b) ω : the angular frequency (角频率), unit: rad/s
 - c) ϕ : the phase (相位), unit: rad or $^\circ$
 - d) $\omega t + \phi$: the argument (辐角), unit: rad or $^\circ$
- *In the sinusoids shown above, the $\phi = 0$.

9.2 Sinusoids (b)

- The period (周期) T and the cyclic frequency (循环频率) f

- a) A periodic function is one function that satisfies $f(t) = f(t + nT)$, for all t and for all integers n .
- b) The period T of the periodic function is the time of one complete cycle or the number of seconds per cycle.
- c) The reciprocal of this quantity is the number of cycles per second, known as the cyclic frequency f of the sinusoid.
- d) Thus it is clear that:

$$f = \frac{1}{T}$$

$$\omega = 2\pi f$$

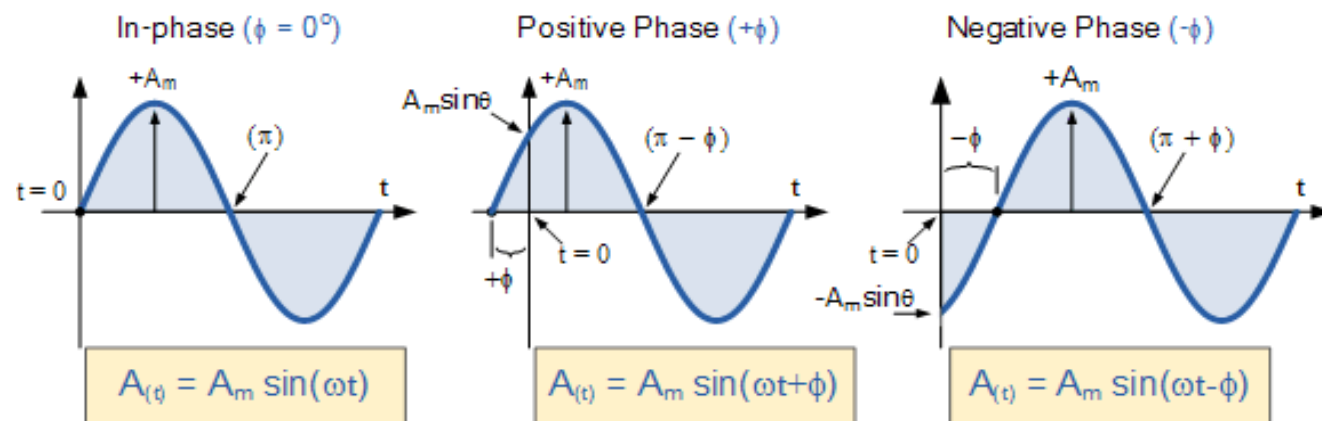
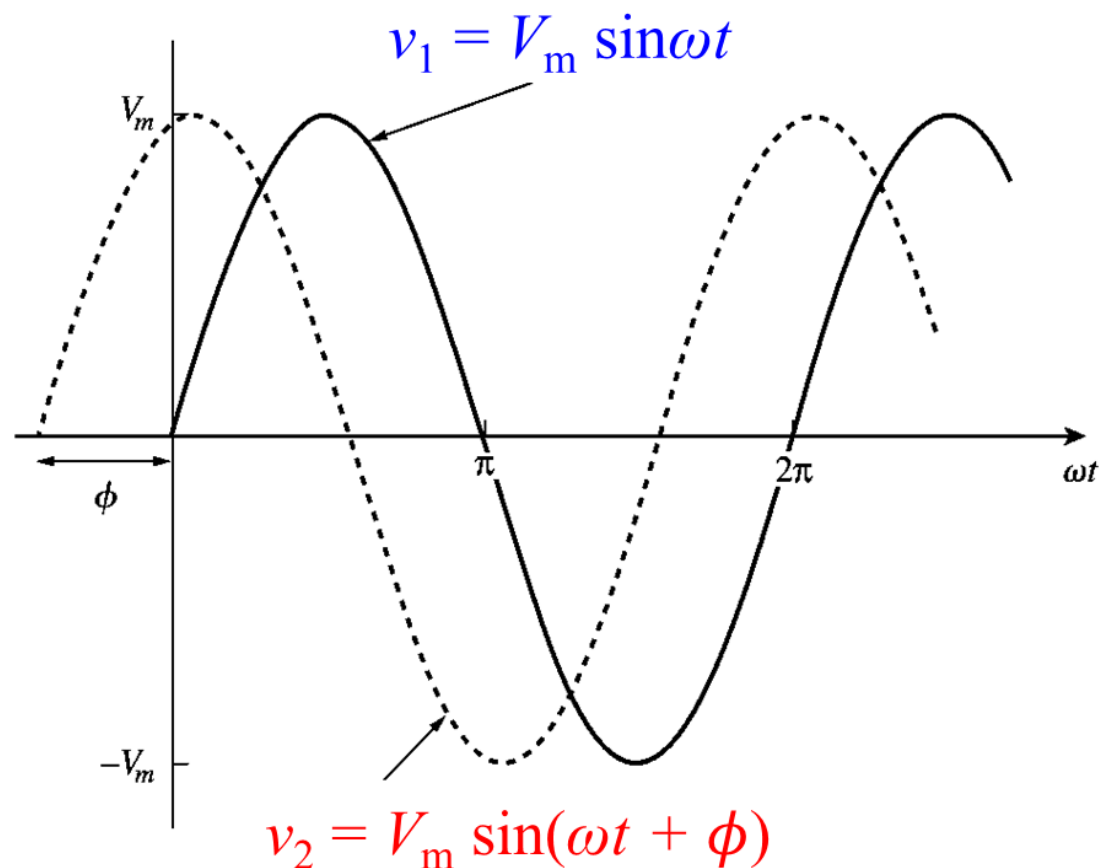
where ω is in rad/s, f is in hertz (Hz).

$$v(t + T) = v(t)$$

$$\begin{aligned} v(t + T) &= V_m \sin \omega(t + T) \\ &= V_m \sin \omega \left(t + \frac{2\pi}{\omega} \right) \\ &= V_m \sin(\omega t + 2\pi) \\ &= V_m \sin \omega t \\ &= v(t) \end{aligned}$$

9.2 Sinusoids (c)

- The phase (相位) ϕ



- By adjusting the phase, the sinusoids can be moved ahead or behind
- The phase of a sinusoid normally means the initial phase.

9.2 Sinusoids (d)

- The phase difference (相位差)

a) We can compare the phases of v_1 and v_2 as they share the ω , while they do not need have the same *amplitude*.

b) $v = V_m \cos(\omega t + \phi_v)$ and $i = I_m \cos(\omega t + \phi_i)$

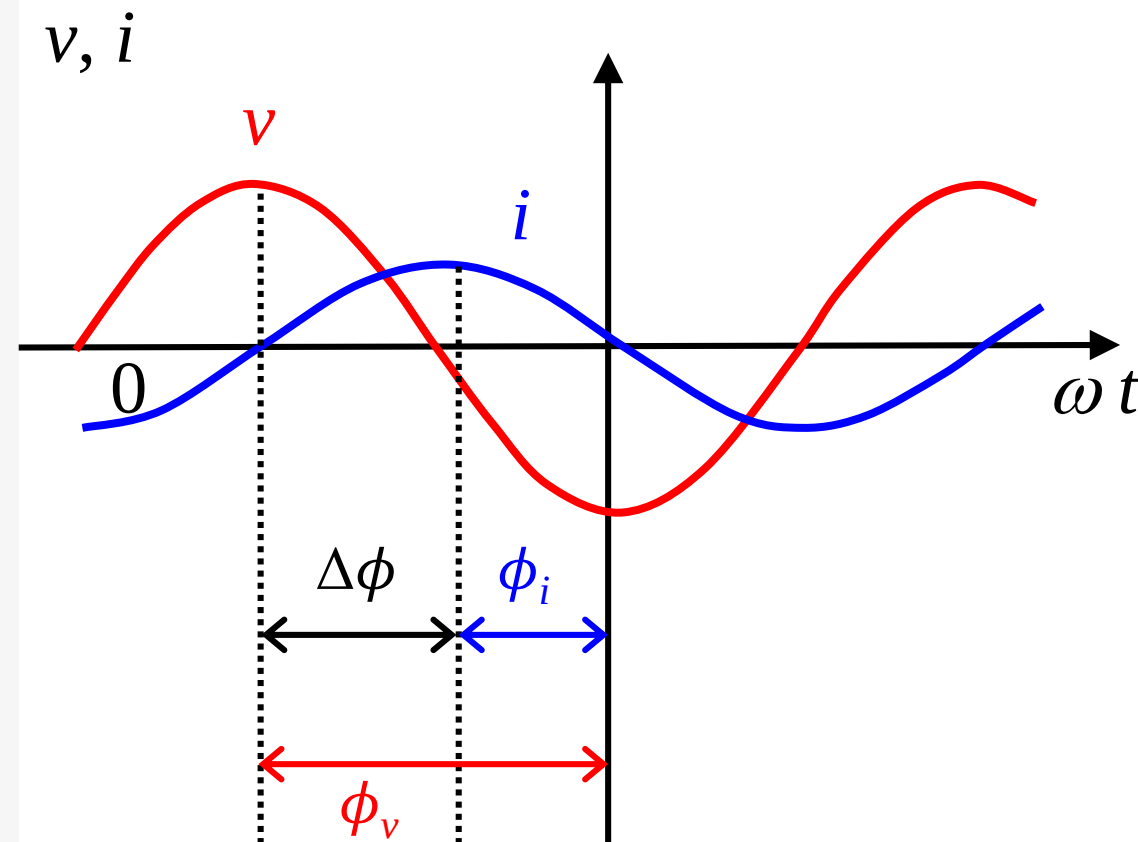
c) Phase difference:

$$\Delta\phi = (\omega t + \phi_v) - (\omega t + \phi_i) = \phi_v - \phi_i$$

d) Generally $|\Delta\phi| \leq \pi$ or $|\Delta\phi| \leq 180^\circ$

e) If $\Delta\phi = \phi_v - \phi_i > 0$, v leads i

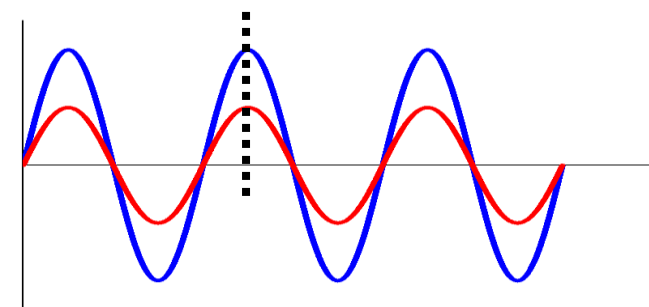
f) If $\Delta\phi = \phi_v - \phi_i < 0$, i leads v



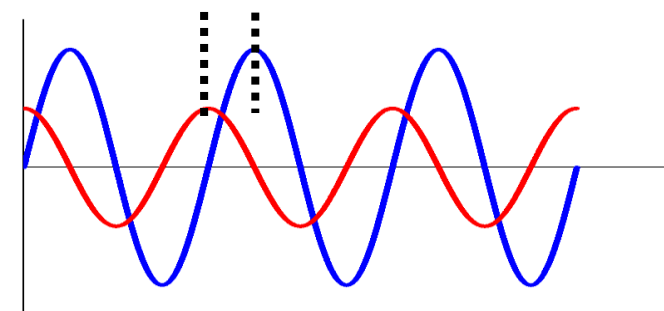
9.2 Sinusoids (e)

- In phase and antiphase

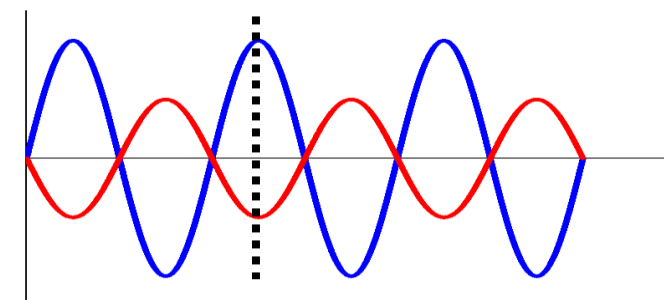
- a) We can compare the phases of v_1 and v_2 as they share the ω , while they do not need have the same *amplitude*.
- b) If $\Delta\phi = 0$ the two sinusoids are **in phase** (同相), they reach to their maximum and minimum at the same time.
- c) If $\Delta\phi \neq 0$, we say that v_1 and v_2 are **out of phase** (异相).
- d) If $\Delta\phi = 180^\circ$, we say v_1 and v_2 are **antiphase** (反相).



In phase
 $\Delta\phi = 0$



Out of phase
 $\Delta\phi \neq 0$



Antiphase
 $\Delta\phi = \pi$

9.2 Sinusoids (f)

- Expression of sinusoids by sin and cos:

a) A sinusoid can be expressed in either sine or cosine form.

b) When comparing two sinusoids, it is expedient to express both as either sine or cosine with positive amplitudes.

c) Using these relationships, we can transform a sinusoid from sine form to cosine form or vice versa.

$$\sin(\omega t \pm 180^\circ) = -\sin \omega t$$

$$\cos(\omega t \pm 180^\circ) = -\cos \omega t$$

$$\sin(\omega t \pm 90^\circ) = \pm \cos \omega t$$

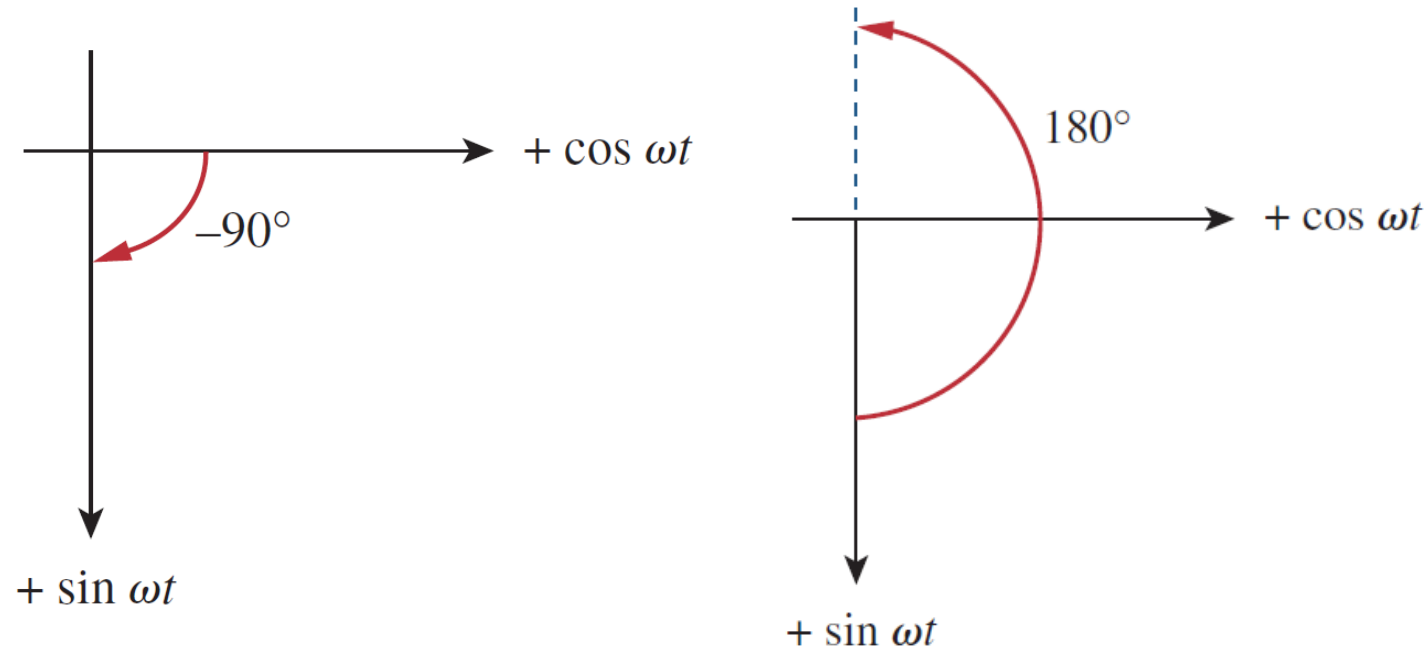
$$\cos(\omega t \pm 90^\circ) = \mp \sin \omega t$$

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

9.2 Sinusoids (g)

- Expression of sinusoids by graphical approaches:



Noted: Angles are measured positively counterclockwise from the horizontal
(逆时针为正角度)

- A graphical approach may be used to relate or compare sinusoids.
- The horizontal axis represents the magnitude of cosine, while the vertical axis (pointing down) denotes the magnitude of sine.

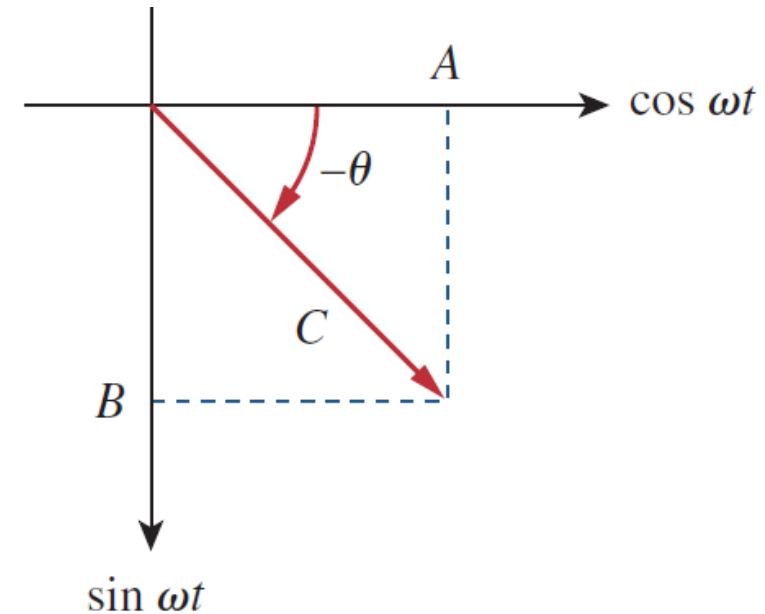
9.2 Sinusoids (h)

- Adding sinusoids by graphical approaches:

When two sinusoids are of the same frequency:

$$A \cos \omega t + B \sin \omega t = C \cos(\omega t - \theta)$$

$$C = \sqrt{A^2 + B^2}, \quad \theta = \tan^{-1} \frac{B}{A}$$



- a) The graphical technique can also be used to add **two sinusoids of the same frequency** when one is in sine form and the other is in cosine form.

9.2 Sinusoids (i)

- Example 9.01:

Example 9.01: Find the amplitude, phase, period and frequency of the following sinusoid:

$$v(t) = 12 \cos(50t + 10^\circ)$$

Solution: The amplitude is $V_m = 12$ V.

The phase is $\phi = 10^\circ$.

The angular frequency is $\omega = 50$ rad/s.

The period $T = \frac{2\pi}{\omega} = \frac{2\pi}{50} = 0.1257$ s.

The frequency is $f = \frac{1}{T} = 7.958$ Hz.

9.2 Sinusoids (j)

- Example 9.02:

Example 9.02: Calculate the phase angle between $v_1 = -10 \cos(\omega t + 50^\circ)$ and $v_2 = 12 \sin(\omega t - 10^\circ)$. State which sinusoid is leading.

$$\sin(\omega t \pm 180^\circ) = -\sin \omega t$$

$$\cos(\omega t \pm 180^\circ) = -\cos \omega t$$

$$\sin(\omega t \pm 90^\circ) = \pm \cos \omega t$$

$$\cos(\omega t \pm 90^\circ) = \mp \sin \omega t$$

Solution I:

$$v_1 = -10 \cos(\omega t + 50^\circ) = 10 \cos(\omega t + 50^\circ - 180^\circ)$$
$$v_1 = 10 \cos(\omega t - 130^\circ) \quad \text{or} \quad v_1 = 10 \cos(\omega t + 230^\circ)$$

$$v_2 = 12 \sin(\omega t - 10^\circ) = 12 \cos(\omega t - 10^\circ - 90^\circ)$$

$$v_2 = 12 \cos(\omega t - 100^\circ)$$

$$v_1 = 10 \cos(\omega t - 130^\circ) \quad v_2 = 12 \cos(\omega t - 100^\circ) \quad v_2 \text{ leads } v_1$$

$$v_1 = 10 \cos(\omega t + 230^\circ) \quad v_2 = 12 \cos(\omega t + 260^\circ) \quad v_2 \text{ leads } v_1$$

9.2 Sinusoids (k)

- Example 9.02:

Example 9.02: Calculate the phase angle between $v_1 = -10 \cos(\omega t + 50^\circ)$ and $v_2 = 12 \sin(\omega t - 10^\circ)$. State which sinusoid is leading.

$$\begin{aligned}\sin(\omega t \pm 180^\circ) &= -\sin \omega t \\ \cos(\omega t \pm 180^\circ) &= -\cos \omega t \\ \sin(\omega t \pm 90^\circ) &= \pm \cos \omega t \\ \cos(\omega t \pm 90^\circ) &= \mp \sin \omega t\end{aligned}$$

Solution II:

$$\begin{aligned}v_1 &= -10 \cos(\omega t + 50^\circ) = 10 \sin(\omega t + 50^\circ - 90^\circ) \\ &= 10 \sin(\omega t - 40^\circ)\end{aligned}$$

$$v_2 = 12 \sin(\omega t - 10^\circ)$$

Therefore, v_2 leads v_1

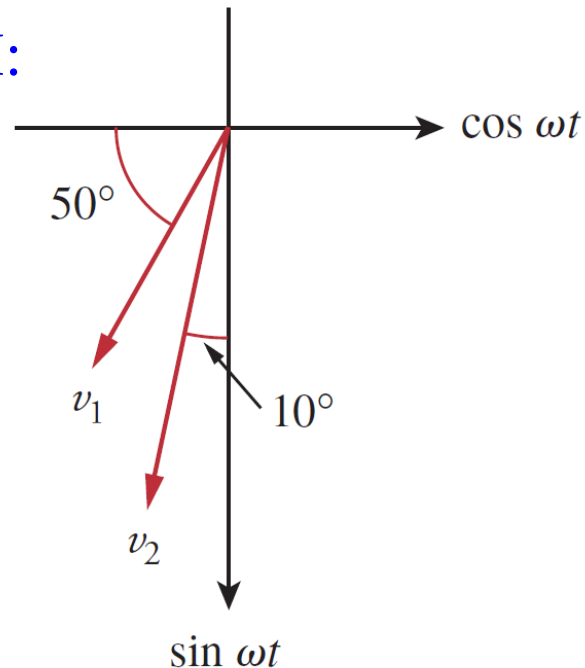
9.2 Sinusoids (I)

- Example 9.02:

Example 9.02: Calculate the phase angle between $v_1 = -10 \cos(\omega t + 50^\circ)$ and $v_2 = 12 \sin(\omega t - 10^\circ)$. State which sinusoid is leading.

$$\begin{aligned}\sin(\omega t \pm 180^\circ) &= -\sin \omega t \\ \cos(\omega t \pm 180^\circ) &= -\cos \omega t \\ \sin(\omega t \pm 90^\circ) &= \pm \cos \omega t \\ \cos(\omega t \pm 90^\circ) &= \mp \sin \omega t\end{aligned}$$

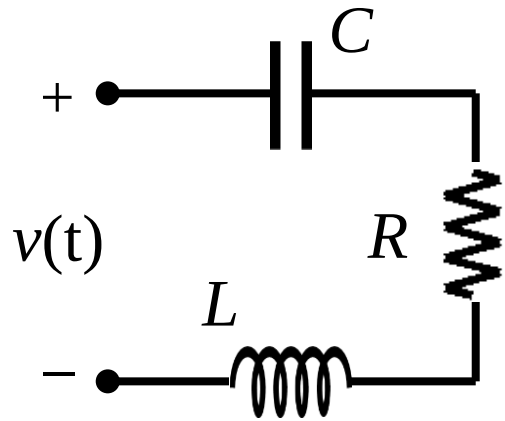
Solution III:



- Express the two sinusoids by the graphical approach.
- Phase difference = $90^\circ - 50^\circ - 10^\circ = 30^\circ$
- v_2 leads v_1 .

9.3 Phasors (a)

- Sinusoid-phasor transformation



$$i(t) = \sqrt{2} I \cos(\omega t + \psi_i)$$

$$u(t) = Ri + L \frac{di}{dt} + \frac{1}{C} \int i dt$$



$$\mathbf{V} = R\mathbf{I} + j\omega L\mathbf{I} + \frac{\mathbf{I}}{j\omega C}$$

Change a time-domain problem into a phasor-domain problem:

- It is much easier to analyze the circuit with the algebraic equation
- The laws and techniques we learned for DC can now be applied to AC circuits.

9.3 Phasors (b)

- A phasor is a complex numbers

$$z = x + jy \quad \text{Rectangular form} \quad \text{直角坐标形式}$$

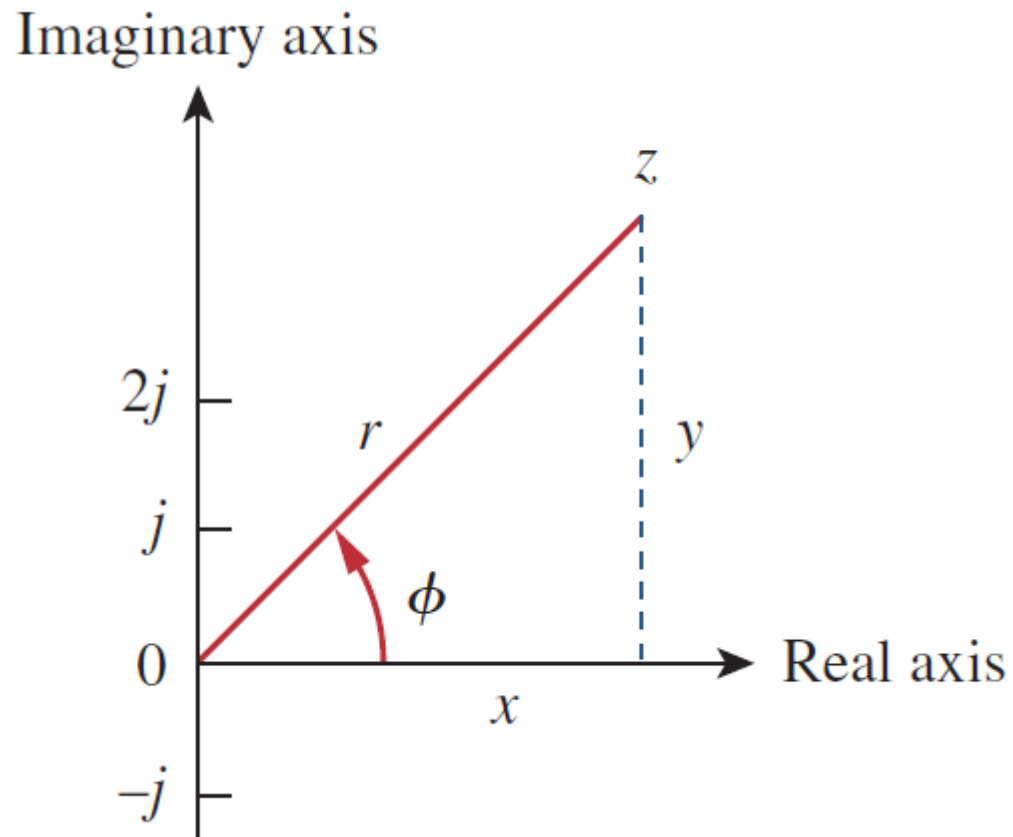
$$z = r \angle \phi \quad \text{Polar form} \quad \text{极坐标形式}$$

$$z = re^{j\phi} \quad \text{Exponential form} \quad \text{指数形式}$$

- Before our discussion on phasors, let's first review some properties of the complex numbers.
- A complex number can take the following forms: where $j = \sqrt{-1}$ being the **imaginary unit**; x and y the real and the imaginary part of z .

9.3 Phasors (c)

- Complex numbers in polar form



$$z = x + jy$$

Rectangular form

$$z = r \angle \phi$$

Polar form

Where, modulus (模) $r = \sqrt{x^2 + y^2}$,

argument (幅角) $\phi = \tan^{-1} \frac{y}{x}$

$$z = x + jy = r \angle \phi = r(\cos \phi + j \sin \phi)$$

9.3 Phasors (d)

- Mathematical operations



a) Addition and subtraction of complex numbers are better performed in rectangular form; multiplication and division are better done in polar form.

$$z_1 = x_1 + jy_1 = r_1 \angle \phi_1 \quad z_2 = x_2 + jy_2 = r_2 \angle \phi_2$$

Addition: $z_1 + z_2 = (x_1 + x_2) + j(y_1 + y_2)$ Subtraction: $z_1 - z_2 = (x_1 - x_2) + j(y_1 - y_2)$

Multiplication: $z_1 z_2 = r_1 r_2 \angle \phi_1 + \phi_2$

Division: $\frac{z_1}{z_2} = \frac{r_1}{r_2} \angle \phi_1 - \phi_2$

Reciprocal: $\frac{1}{z} = \frac{1}{r} \angle -\phi \rightarrow \frac{1}{j} = -j$

Square root: $\sqrt{z} = \sqrt{r} \angle \phi/2$

Complex conjugate: $z^* = x - jy = r \angle -\phi = re^{-j\phi}$

9.3 Phasors (e)

- Phasors (相量)

$$v(t) = V \cos(\omega t + \phi), \text{ let } f(x) = V e^{j(\omega t + \phi)}$$

$$\cos \phi = \operatorname{Re}(e^{j\phi})$$

$$\sin \phi = \operatorname{Im}(e^{j\phi})$$

From the Euler formula, $e^{\pm j\phi} = \cos \phi \pm j \sin \phi$, we have:

$$f(x) = V \cos(\omega t + \phi) + jV \sin(\omega t + \phi) = v(t) + jV \sin(\omega t + \phi)$$

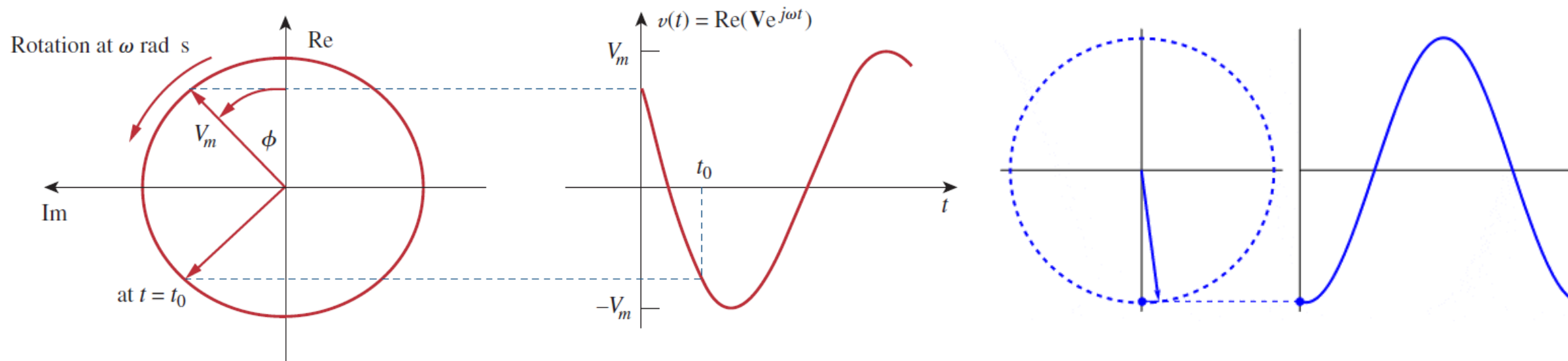
$$\text{So } v(t) = \operatorname{Re}(f(x)) = \operatorname{Re}(V e^{j(\omega t + \phi)}), \text{ or } v(t) = \operatorname{Re}(V_m e^{j\phi} e^{j\omega t}) = \operatorname{Re}(\mathbf{V} e^{j\omega t})$$

Where $\mathbf{V} = V_m e^{j\phi} = V_m \angle \phi$

is called the phasor (相量) representation of the original sinusoid $v(t)$.

9.3 Phasors (f)

- Phasors (相量)



- a) A phasor is a complex representation of the magnitude and phase of a sinusoid.
- b) The sinusoid may be regarded as a rotating phasor.

9.3 Phasors (g)

- Sinusoid-phasor transformation

$$\begin{array}{ccc} v(t) = V_m \cos(\omega t + \phi) & \Leftrightarrow & \mathbf{V} = V_m \angle \phi \\ \text{(Time-domain representation)} & & \text{(Phasor-domain representation)} \end{array}$$

TABLE 9.1

Sinusoid-phasor transformation.

Time domain representation	Phasor domain representation
$V_m \cos(\omega t + \phi)$	$V_m \angle \phi$
$V_m \sin(\omega t + \phi)$	$V_m \angle \phi - 90^\circ$
$I_m \cos(\omega t + \theta)$	$I_m \angle \theta$
$I_m \sin(\omega t + \theta)$	$I_m \angle \theta - 90^\circ$

$$\sin(\omega t \pm 90^\circ) = \pm \cos \omega t$$

$$\cos(\omega t \pm 90^\circ) = \mp \sin \omega t$$

9.3 Phasors (h)

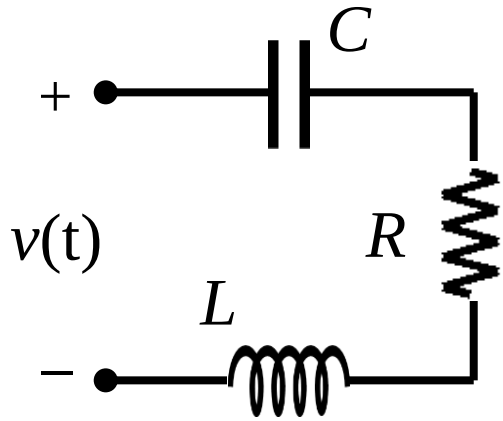
- Sinusoid-phasor transformation

$$\begin{array}{ccc} v(t) = V_m \cos(\omega t + \phi) & \Leftrightarrow & \mathbf{V} = V_m \angle \phi \\ \text{(Time-domain} & & \text{(Phasor-domain} \\ \text{representation)} & & \text{representation)} \end{array}$$

- To get the phasor representation of a sinusoid, we express it in cosine form and take the magnitude and phase.
- Given a phasor, we obtain the time domain representation as the cosine function with the same magnitude as the phasor and the argument as plus the phase of the phasor.
- The idea of expressing information in alternate domains is fundamental to all areas of engineering.

9.3 Phasors (i)

- Sinusoid-phasor transformation



$$i(t) = \sqrt{2} I \cos(\omega t + \psi_i)$$

$$u(t) = Ri + L \frac{di}{dt} + \frac{1}{C} \int i dt$$



$$\mathbf{V} = R\mathbf{I} + j\omega L\mathbf{I} + \frac{\mathbf{I}}{j\omega C}$$

Change a time-domain problem into a phasor-domain problem:

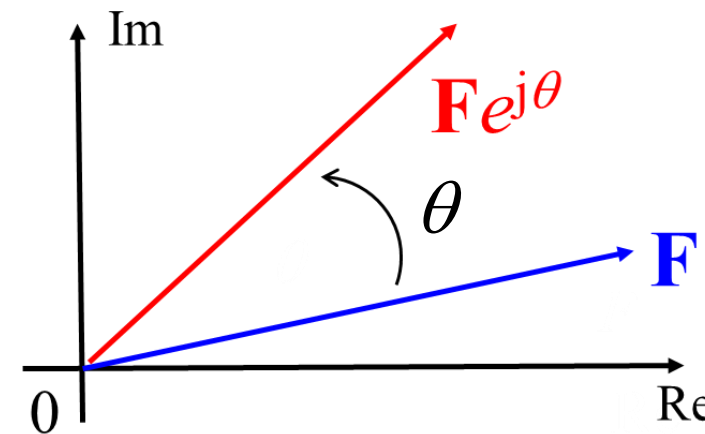
- It is much easier to analyze the circuit with the algebraic equation
- The laws and techniques we learned for DC can now be applied to AC circuits.

9.3 Phasors (j)

- Special phasors: rotating factors (旋转因子)

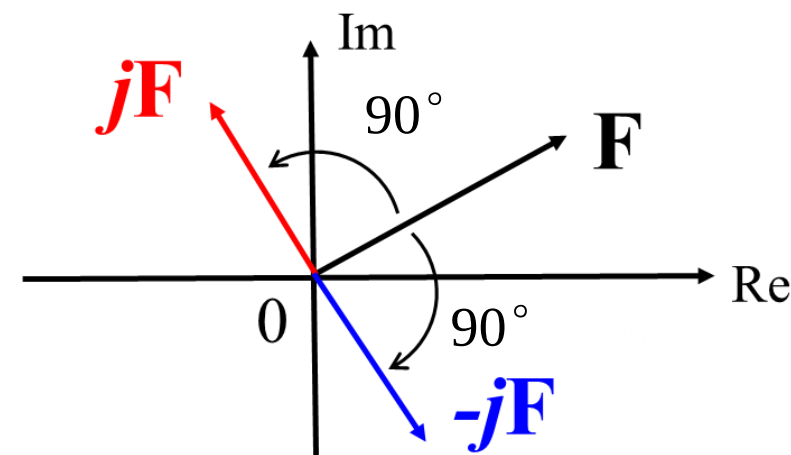
For a complex number: $e^{j\theta} = \cos\theta + j\sin\theta = 1 \angle \theta$

This can be used to rotate a phasor, $\mathbf{F}e^{j\theta}$



$\pm j$ is a special rotating factor.

$$e^{\pm j90^\circ} = \cos 90^\circ \pm j\sin 90^\circ = 0 \pm j = \pm j$$



9.3 Phasors (k)

• Example 9.03

Transform these sinusoids to phasors:

(a) $i = 6 \cos(50t - 40^\circ) \text{ A}$

(b) $v = -4 \sin(30t + 50^\circ) \text{ V}$

Solution:

(a) $i = 6 \cos(50t - 40^\circ)$ has the phasor

$$\mathbf{I} = 6 \angle -40^\circ \text{ A}$$

(b) Since $-\sin A = \cos(A + 90^\circ)$,

$$\begin{aligned} v &= -4 \sin(30t + 50^\circ) = 4 \cos(30t + 50^\circ + 90^\circ) \\ &= 4 \cos(30t + 140^\circ) \text{ V} \end{aligned}$$

The phasor form of v is

$$\mathbf{V} = 4 \angle 140^\circ \text{ V}$$

9.3 Phasors (I)

- Example 9.04

Find the sinusoids represented by these phasors: (a) $\mathbf{I} = -3 + j4$ A (b) $\mathbf{V} = j8e^{-j20^\circ}$ V

Solution:

(a) $\mathbf{I} = -3 + j4 = 5\angle 126.87^\circ$. Transforming this to the time domain gives

$$i(t) = 5 \cos(\omega t + 126.87^\circ) \text{ A}$$

(b) Since $j = 1\angle 90^\circ$,

$$\begin{aligned}\mathbf{V} &= j8\angle -20^\circ = (1\angle 90^\circ)(8\angle -20^\circ) \\ &= 8\angle 90^\circ - 20^\circ = 8\angle 70^\circ \text{ V}\end{aligned}$$

Converting this to the time domain gives

$$v(t) = 8 \cos(\omega t + 70^\circ) \text{ V}$$

9.3 Phasors (m)

- Sinusoid-phasor transformation

$$\sin(\omega t \pm 90^\circ) = \pm \cos \omega t$$

$$\cos(\omega t \pm 90^\circ) = \mp \sin \omega t$$

$$v(t) = \text{Re}(\mathbf{V}e^{j\omega t}) = V_m \cos(\omega t + \phi)$$

$$\begin{aligned} \Rightarrow \frac{dv}{dt} &= -\omega V_m \sin(\omega t + \phi) = \omega V_m \cos(\omega t + \phi + 90^\circ) \\ &= \text{Re}(\omega V_m e^{j\omega t} e^{j\phi} e^{j90^\circ}) = \text{Re}(j\omega \mathbf{V}e^{j\omega t}) \end{aligned}$$

$$\frac{dv}{dt}$$

(Time domain)



$$j\omega \mathbf{V}$$

(Phasor domain)

Differentiating a sinusoid is equivalent to multiplying its corresponding phasor by $j\omega$.

9.3 Phasors (n)

- Sinusoid-phasor transformation

Differentiating a sinusoid is equivalent to multiplying its corresponding phasor by $j\omega$.

$$\frac{dv}{dt} \quad \Leftrightarrow \quad j\omega V$$

(Time domain) (Phasor domain)

Integrating a sinusoid is equivalent to dividing its corresponding phasor by $j\omega$.

$$\int v dt \quad \Leftrightarrow \quad \frac{V}{j\omega}$$

(Time domain) (Phasor domain)

Adding sinusoids of the same frequency is equivalent to adding their corresponding phasors.

9.3 Phasors (p)

- Example 9.05

Example 9.05: Given $i_1(t) = 4 \cos(\omega t + 30^\circ)$ and $i_2(t) = 5 \sin(\omega t - 20^\circ)$, find their sum.

Solution:

Here is an important use of phasors—for summing sinusoids of the same frequency. Current $i_1(t)$ is in the standard form. Its phasor is

$$\mathbf{I}_1 = 4 \angle 30^\circ$$

We need to express $i_2(t)$ in cosine form. The rule for converting sine to cosine is to subtract 90° . Hence,

$$i_2 = 5 \cos(\omega t - 20^\circ - 90^\circ) = 5 \cos(\omega t - 110^\circ)$$

and its phasor is

$$\mathbf{I}_2 = 5 \angle -110^\circ$$

$$\begin{aligned}\sin(A \pm B) &= \sin A \cos B \pm \cos A \sin B \\ \cos(A \pm B) &= \cos A \cos B \mp \sin A \sin B\end{aligned}$$

9.3 Phasors (q)

- Example 9.06

Example 9.06: Given $i_1(t) = 4 \cos(\omega t + 30^\circ)$ and $i_2(t) = 5 \sin(\omega t - 20^\circ)$, find their sum.

If we let $i = i_1 + i_2$, then

$$\begin{aligned} \mathbf{I} &= \mathbf{I}_1 + \mathbf{I}_2 = 4\angle 30^\circ + 5\angle -110^\circ \\ &= 4\cos 30^\circ + j4\sin 30^\circ + 5\cos(-110^\circ) + j5\sin(-110^\circ) \\ &= 3.464 + j2 - 1.71 - j4.698 = 1.754 - j2.698 \\ &= 3.218\angle -56.97^\circ \text{ A} \end{aligned}$$

9.3 Phasors (x)

- Example 9.07

Using the phasor approach, determine the current $i(t)$ in a circuit described by the integrodifferential equation

$$4i + 8 \int i dt - 3 \frac{di}{dt} = 50 \cos(2t + 75^\circ)$$

Solution: Transform the equation from time domain to phasor domain:

$$4\mathbf{I} + \frac{8\mathbf{I}}{j\omega} - 3j\omega\mathbf{I} = 50\angle 75^\circ$$

Because $\omega = 2$ & $1/j = -j$:

$$\mathbf{I}(4 - j4 - j6) = 50\angle 75^\circ$$

$$\mathbf{I} = \frac{50\angle 75^\circ}{4 - j10} = \frac{50\angle 75^\circ}{10.77\angle -68.2^\circ} = 4.642\angle 143.2^\circ \text{ A}$$

Lesson 9 Sinusoids and Phasors

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9.4 Impedances and admittances (a)

- Ohm's law in phasor domain - resistors

- For the resistor, the voltage and current are related via Ohm's law in time domain.
- As such, the voltage and current are in phase with each other.

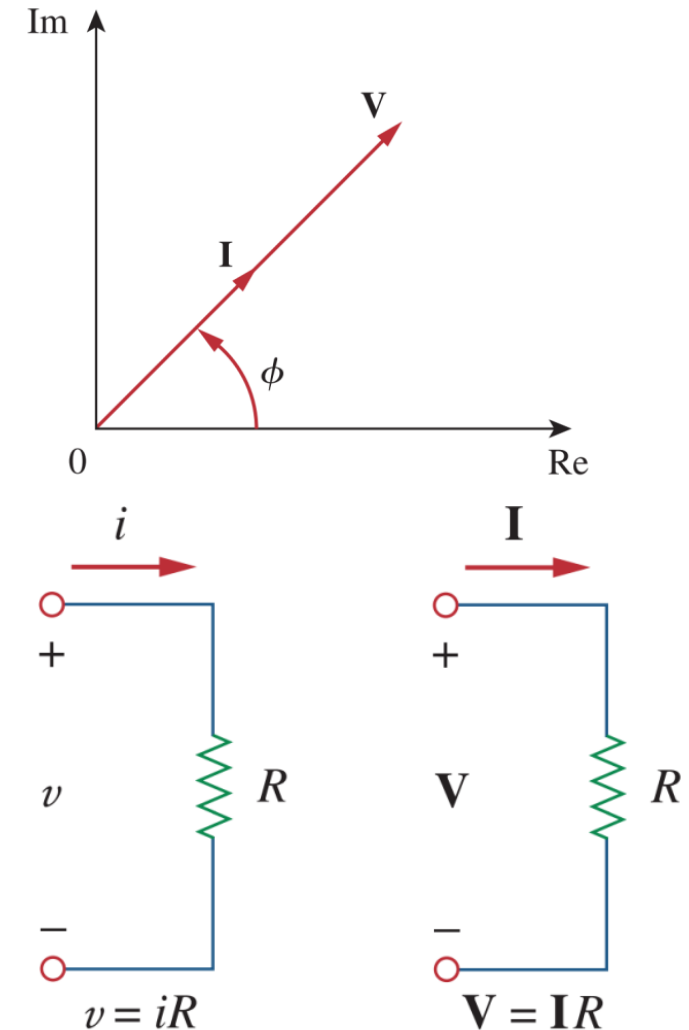
$$v = iR = RI_m \cos(\omega t + \phi)$$

The phasor form of this voltage is

$$\mathbf{V} = RI_m \angle \phi$$

But the phasor representation of the current is $\mathbf{I} = I_m \angle \phi$. Hence,

$$\mathbf{V} = R\mathbf{I}$$



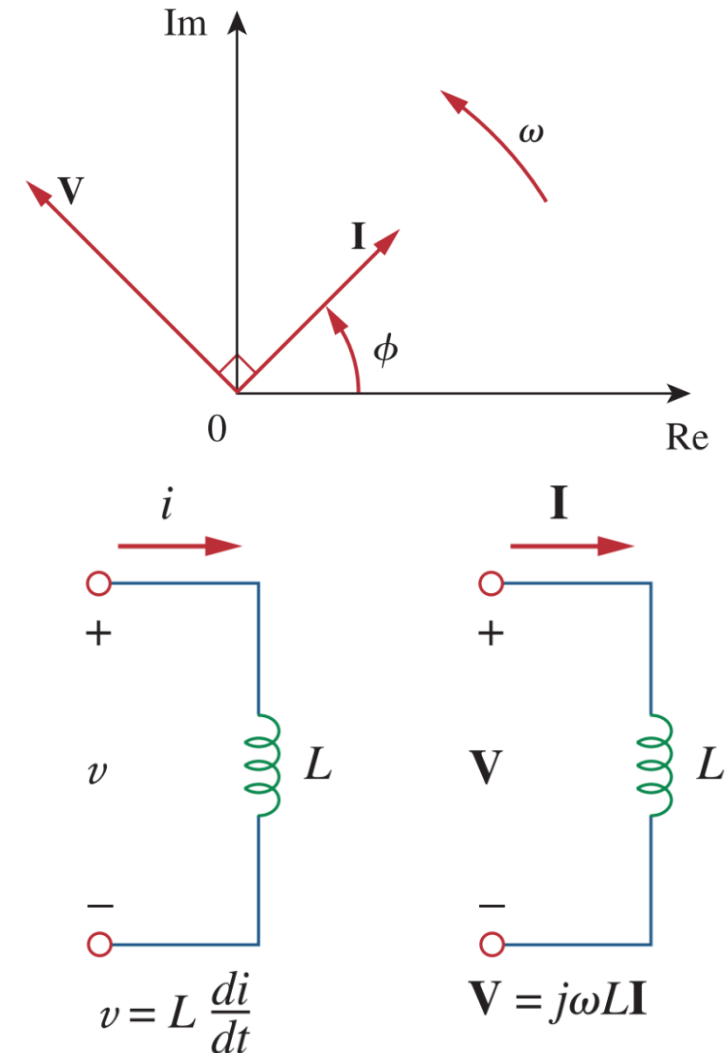
9.4 Impedances and admittances (b)

- Ohm's law in phasor domain - inductors

$$v = L \frac{di}{dt} = -\omega L I_m \sin(\omega t + \phi) = \omega L I_m \cos(\omega t + \phi + 90^\circ)$$

$$\mathbf{V} = j\omega L \mathbf{I}$$

- Inductors on the other hand have a phase shift between the voltage and current.
- In this case, **the voltage leads the current by 90°** . Or one says the current lags the voltage, which is the standard convention.
- This is represented on the phasor diagram by a positive phase angle between the voltage and current.

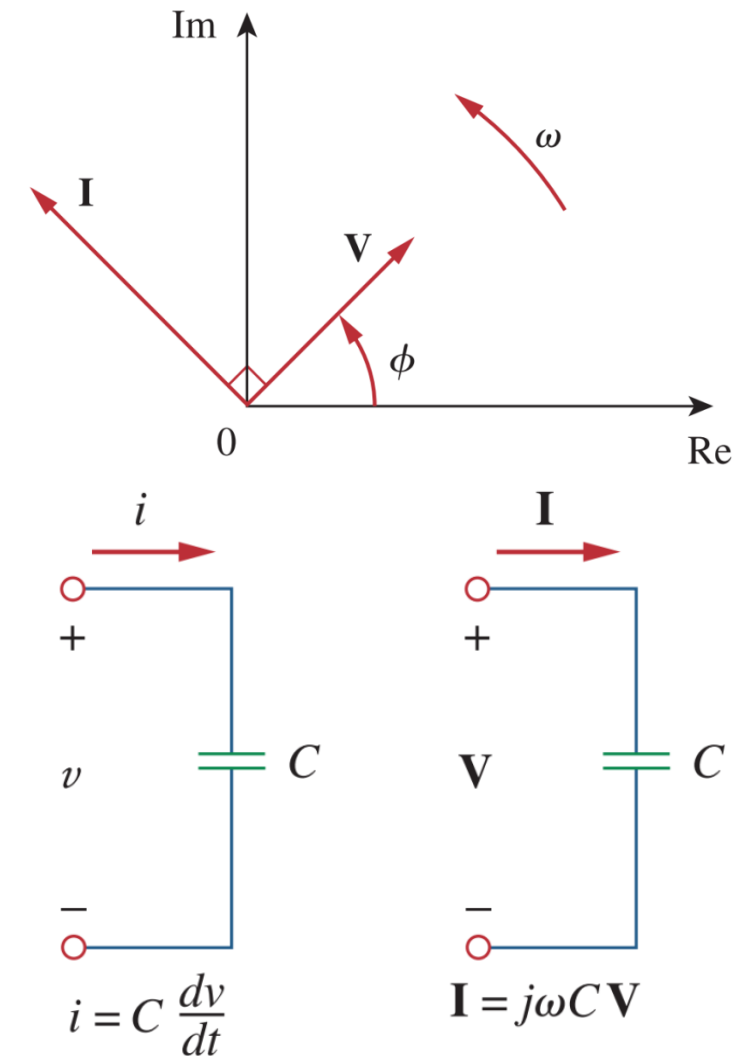


9.4 Impedances and admittances (c)

- Ohm's law in phasor domain - capacitors

$$i = C \frac{dv}{dt} \quad \mathbf{I} = j\omega C \mathbf{V} \quad \Rightarrow \quad \mathbf{V} = \frac{\mathbf{I}}{j\omega C}$$

- Capacitors have the opposite phase relationship as compared to the inductors.
- In this case, the current leads the voltage by 90° .
- In a phasor diagram, this corresponds to a negative phase angle between the voltage and current.



9.4 Impedances and admittances (d)

- CIVIL

CIVIL

In a **C**apacitor **I** leads **V**, but **V** leads **I** in an inductor (**L**).

9.4 Impedances and admittances (e)

- Ohm's law in phasor domain

TABLE 9.2

Summary of voltage-current relationships.

Element	Time domain	Frequency domain
R	$v = Ri$	$\mathbf{V} = R\mathbf{I}$
L	$v = L \frac{di}{dt}$	$\mathbf{V} = j\omega L\mathbf{I}$
C	$i = C \frac{dv}{dt}$	$\mathbf{V} = \frac{\mathbf{I}}{j\omega C}$

9.4 Impedances and admittances (f)

- Ohm's law in phasor domain

Resistor	Inductor	Capacitor
$\mathbf{V} = R\mathbf{I},$	$\mathbf{V} = j\omega L\mathbf{I},$	$\mathbf{V} = \frac{\mathbf{I}}{j\omega C}$
$\frac{\mathbf{V}}{\mathbf{I}} = R,$	$\frac{\mathbf{V}}{\mathbf{I}} = j\omega L,$	$\frac{\mathbf{V}}{\mathbf{I}} = \frac{1}{j\omega C}$

a) From these three expressions, we obtain ohm's law in phasor form for any type of element as:

$$\mathbf{Z} = \frac{\mathbf{V}}{\mathbf{I}} \quad \text{or} \quad \mathbf{V} = \mathbf{Z}\mathbf{I}$$

b) $\mathbf{Z}$ is a frequency-dependent quantity known as **impedance (阻抗)**, measured in ohms. The **admittance (Y, 导纳)** is the reciprocal of impedance, measured in siemens.

9.4 Impedances and admittances (g)

- Impedances

TABLE 9.3

Impedances and admittances of passive elements.

Element	Impedance	Admittance
R	$Z = R$	$Y = \frac{1}{R}$
L	$Z = j\omega L$	$Y = \frac{1}{j\omega L}$
C	$Z = \frac{1}{j\omega C}$	$Y = j\omega C$

Resistance
电阻

Reactance
电抗

$$\mathbf{Z} = R + jX = |\mathbf{Z}| \angle \theta$$

$$|\mathbf{Z}| = \sqrt{R^2 + X^2}, \quad \theta = \tan^{-1} \frac{X}{R}$$

$$R = |\mathbf{Z}| \cos \theta, \quad X = |\mathbf{Z}| \sin \theta$$

- If x is positive, we say the impedance is inductive (感性阻抗或滞后阻抗), i lags v .
- If x is negative, we say the impedance is capacitive (容性阻抗或超前阻抗), i leads v .

9.4 Impedances and admittances (h)

- Impedances and reactance

$$\mathbf{Z} = R + jX = |\mathbf{Z}| \angle \theta$$

Element	Impedance $\mathbf{Z}$	Resistance R	Reactance X
R	R	R	-
L	$j\omega L$	-	ωL
C	$1/j\omega C = -j/\omega C$	-	$-1/\omega C$

9.4 Impedances and admittances (i)

- Admittance

TABLE 9.3

Impedances and admittances of passive elements.

Element	Impedance	Admittance
R	$Z = R$	$Y = \frac{1}{R}$
L	$Z = j\omega L$	$Y = \frac{1}{j\omega L}$
C	$Z = \frac{1}{j\omega C}$	$Y = j\omega C$

Conductance
电导

Susceptance
电纳

$$Y = G + jB$$

$$G + jB = \frac{1}{R + jX} \cdot \frac{R - jX}{R - jX} = \frac{R - jX}{R^2 + X^2}$$

$$G = \frac{R}{R^2 + X^2}, \quad B = -\frac{X}{R^2 + X^2}$$

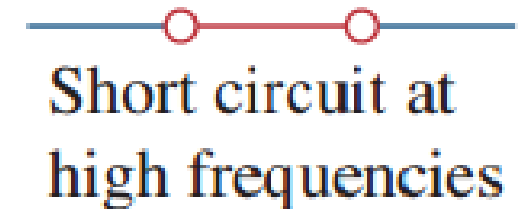
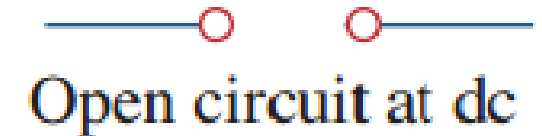
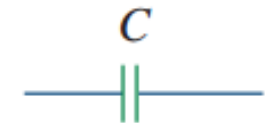
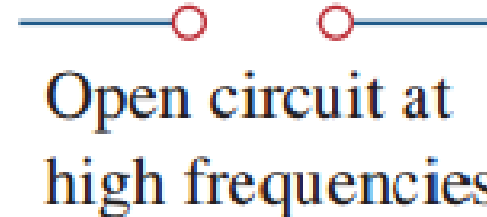
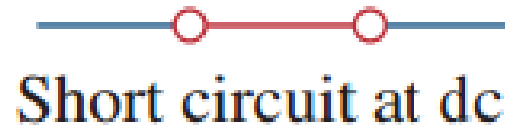
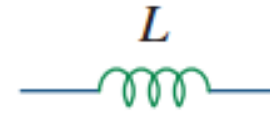
9.4 Impedances and admittances (j)

- Impedance can depend on the frequency

TABLE 9.3

Impedances and admittances of passive elements.

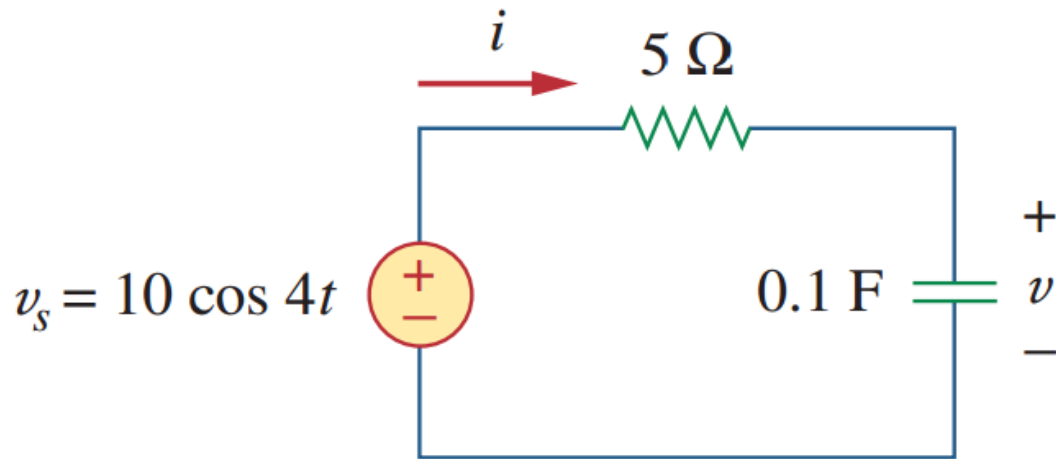
Element	Impedance	Admittance
R	$Z = R$	$Y = \frac{1}{R}$
L	$Z = j\omega L$	$Y = \frac{1}{j\omega L}$
C	$Z = \frac{1}{j\omega C}$	$Y = j\omega C$



9.4 Impedances and admittances (k)

- Example 9.08

Example 9.08: Find $v(t)$ and $i(t)$ in the circuit.



Solution:

From the voltage source $10 \cos 4t$, $\omega = 4$,

$$\mathbf{V}_s = 10 \angle 0^\circ \text{ V}$$

The impedance is

$$\mathbf{Z} = 5 + \frac{1}{j\omega C} = 5 + \frac{1}{j4 \times 0.1} = 5 - j2.5 \Omega$$

Hence the current

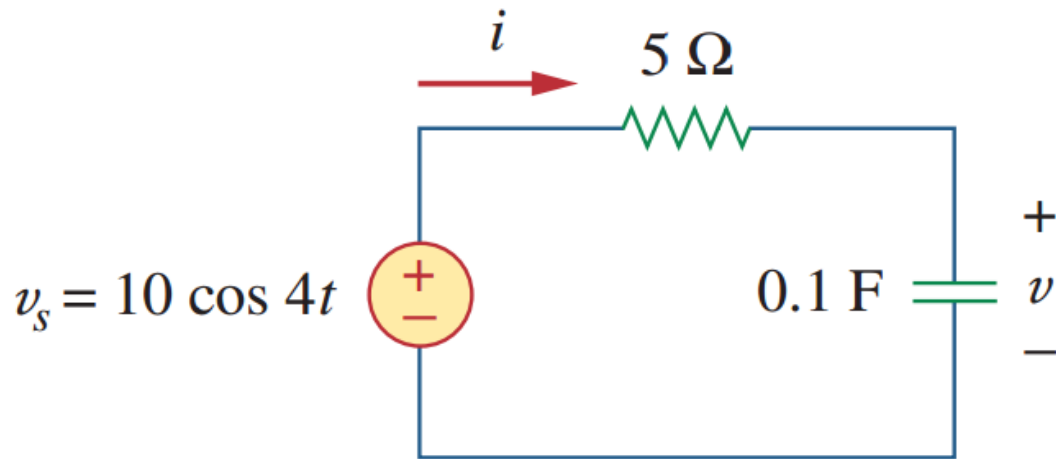
$$\begin{aligned} \mathbf{I} &= \frac{\mathbf{V}_s}{\mathbf{Z}} = \frac{10 \angle 0^\circ}{5 - j2.5} = \frac{10(5 + j2.5)}{5^2 + 2.5^2} \\ &= 1.6 + j0.8 = 1.789 \angle 26.57^\circ \text{ A} \end{aligned}$$

$$i(t) = 1.789 \cos(4t + 26.57^\circ) \text{ A}$$

9.4 Impedances and admittances (I)

- Example 9.08

Example 9.08: Find $v(t)$ and $i(t)$ in the circuit.



Solution: the voltage across the capacitor is:

$$\begin{aligned} \mathbf{V} &= \mathbf{I} \mathbf{Z}_C = \frac{\mathbf{I}}{j\omega C} = \frac{1.789 \angle 26.57^\circ}{j4 \times 0.1} \\ &= \frac{1.789 \angle 26.57^\circ}{0.4 \angle 90^\circ} = 4.47 \angle -63.43^\circ \text{ V} \end{aligned}$$

$$v(t) = 4.47 \cos(4t - 63.43^\circ) \text{ V}$$

$$i(t) = 1.789 \cos(4t + 26.57^\circ) \text{ A}$$

Notice that $i(t)$ leads $v(t)$ by 90° as expected.

Lesson 9 Sinusoids and Phasors

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9.5 KVL and KCL in frequency domain (a)

- KVL

$$v_1 + v_2 + \cdots + v_n = 0 \quad \Rightarrow \quad V_{m1} \cos(\omega t + \theta_1) + V_{m2} \cos(\omega t + \theta_2) + \cdots + V_{mn} \cos(\omega t + \theta_n) = 0$$

This can be written as

$$\operatorname{Re}(V_{m1}e^{j\theta_1}e^{j\omega t}) + \operatorname{Re}(V_{m2}e^{j\theta_2}e^{j\omega t}) + \cdots + \operatorname{Re}(V_{mn}e^{j\theta_n}e^{j\omega t}) = 0$$

or

$$\operatorname{Re}[(V_{m1}e^{j\theta_1} + V_{m2}e^{j\theta_2} + \cdots + V_{mn}e^{j\theta_n})e^{j\omega t}] = 0$$

If we let $\mathbf{V}_k = V_{mk}e^{j\theta_k}$, then

$$\operatorname{Re}[(\mathbf{V}_1 + \mathbf{V}_2 + \cdots + \mathbf{V}_n)e^{j\omega t}] = 0$$

Since $e^{j\omega t} \neq 0$,

$$\mathbf{V}_1 + \mathbf{V}_2 + \cdots + \mathbf{V}_n = 0$$

9.5 KVL and KCL in frequency domain (b)

- KCL

By following a similar procedure, we can show that Kirchhoff's current law holds for phasors. If we let $i_1, i_2, \dots, i_n$ be the current leaving or entering a closed surface in a network at time t , then

$$i_1 + i_2 + \dots + i_n = 0$$

If $\mathbf{I}_1, \mathbf{I}_2, \dots, \mathbf{I}_n$ are the phasor forms of the sinusoids $i_1, i_2, \dots, i_n$, then

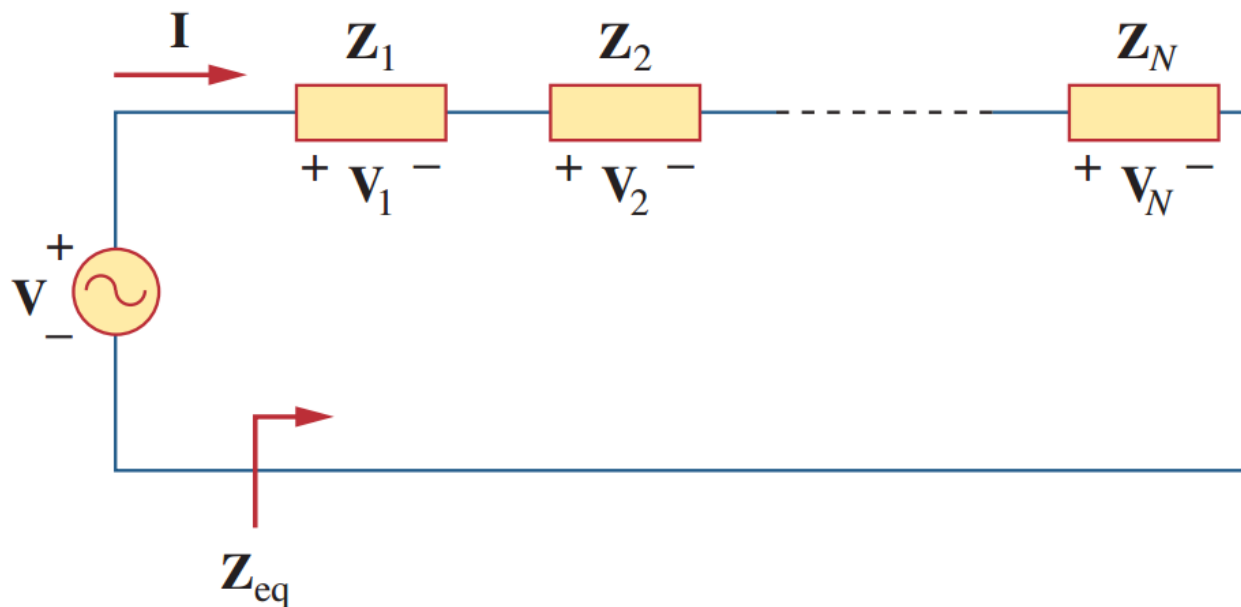
$$\mathbf{I}_1 + \mathbf{I}_2 + \dots + \mathbf{I}_n = 0$$

which is Kirchhoff's current law in the frequency domain.

Once we have shown that both KVL and KCL hold in the frequency domain, it is easy to do many things, such as impedance combination, nodal and mesh analyses, superposition, and source transformation.

9.5 KVL and KCL in frequency domain (c)

- Impedances in series (voltage dividing):

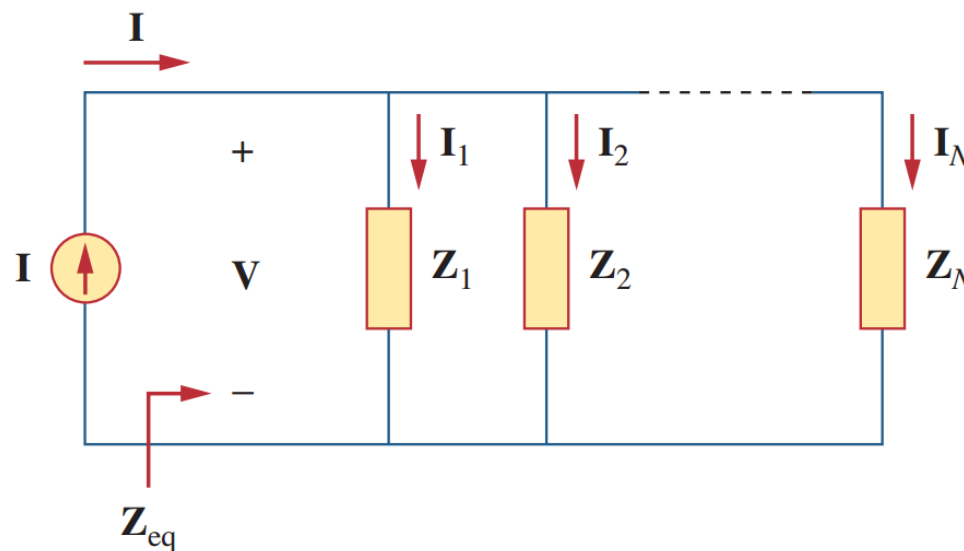


$$V = V_1 + V_2 + \cdots + V_N = I(Z_1 + Z_2 + \cdots + Z_N)$$

$$Z_{eq} = \frac{V}{I} = Z_1 + Z_2 + \cdots + Z_N$$

9.5 KVL and KCL in frequency domain (d)

- Impedances in parallel (current dividing):

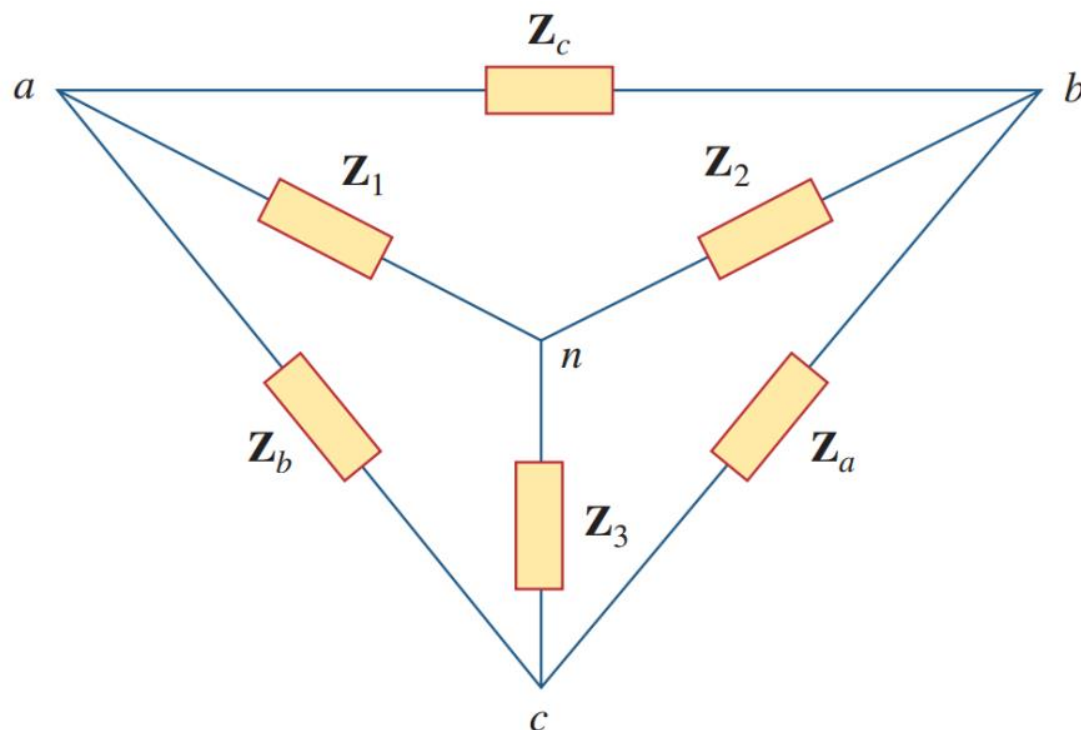


$$I = I_1 + I_2 + \cdots + I_N = V \left(\frac{1}{Z_1} + \frac{1}{Z_2} + \cdots + \frac{1}{Z_N} \right)$$

$$\frac{1}{Z_{eq}} = \frac{I}{V} = \frac{1}{Z_1} + \frac{1}{Z_2} + \cdots + \frac{1}{Z_N}$$

9.5 KVL and KCL in frequency domain (e)

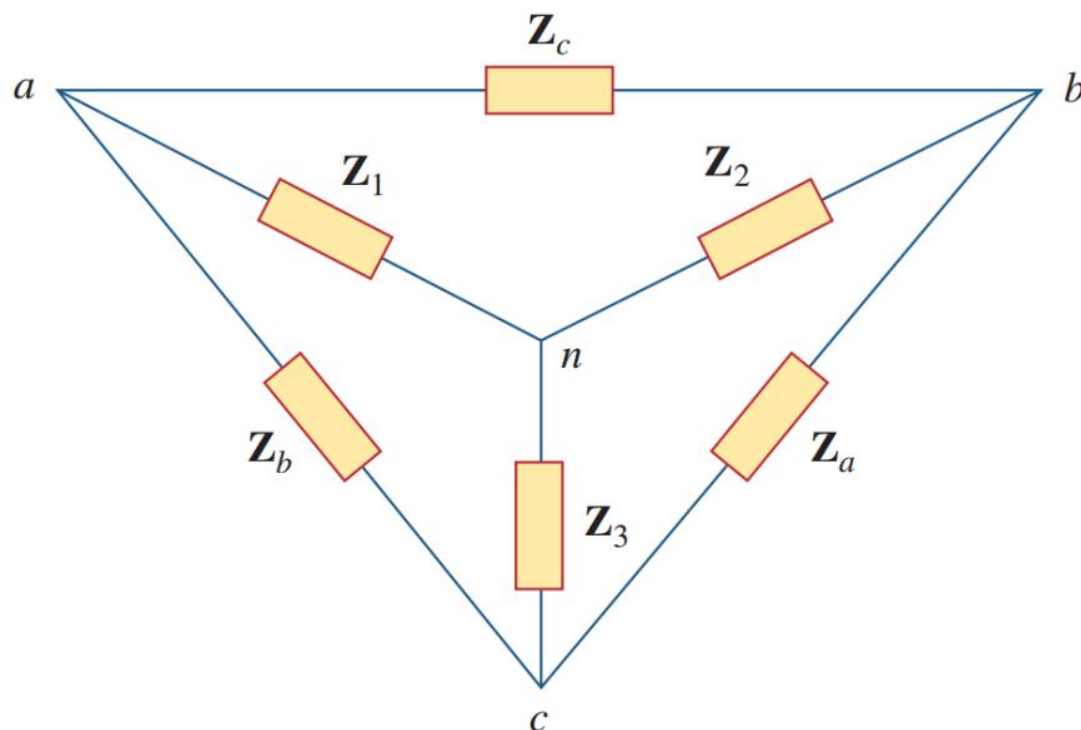
- Δ to Y transformation:



$$\begin{aligned} Z_1 &= \frac{Z_b Z_c}{Z_a + Z_b + Z_c} \\ Z_2 &= \frac{Z_c Z_a}{Z_a + Z_b + Z_c} \\ Z_3 &= \frac{Z_a Z_b}{Z_a + Z_b + Z_c} \end{aligned}$$

9.5 KVL and KCL in frequency domain (f)

- Y to Δ transformation:



$$Z_a = \frac{Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1}{Z_1}$$

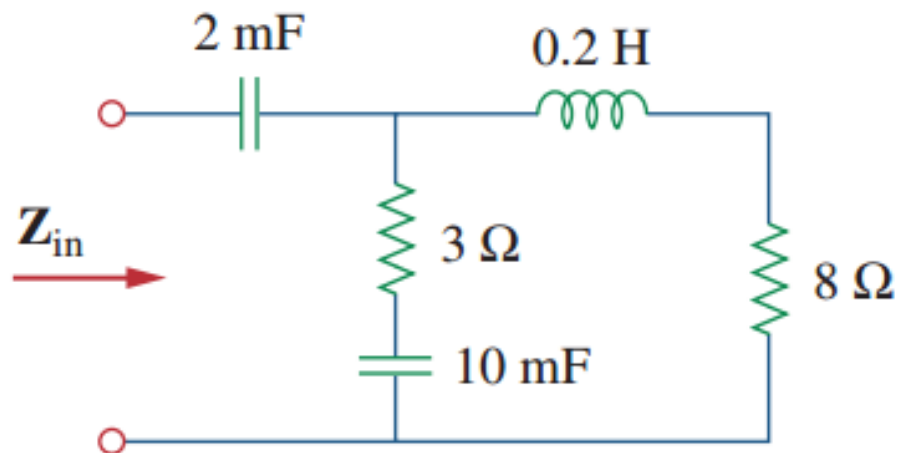
$$Z_b = \frac{Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1}{Z_2}$$

$$Z_c = \frac{Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1}{Z_3}$$

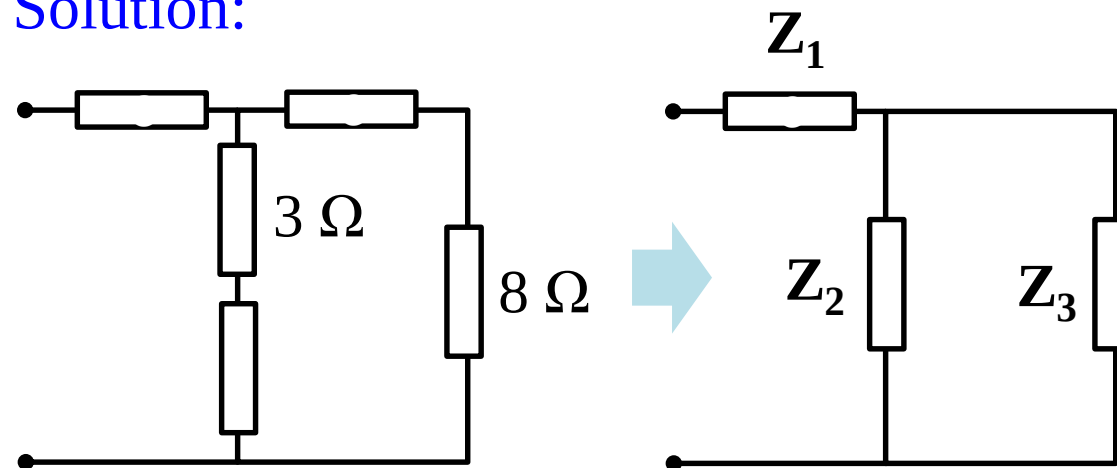
9.5 KVL and KCL in frequency domain (g)

- Example 9.09:

Example 9.09: find the input impedance in the circuit, assume that $\omega = 50 \text{ rad/s}$.



Solution:



$$Z_1 = \frac{1}{j\omega C} = \frac{1}{j50 \times 2 \times 10^{-3}} = -j10 \Omega$$

$$Z_2 = 3 + \frac{1}{j\omega C} = 3 + \frac{1}{j50 \times 10 \times 10^{-3}} = (3 - j2) \Omega$$

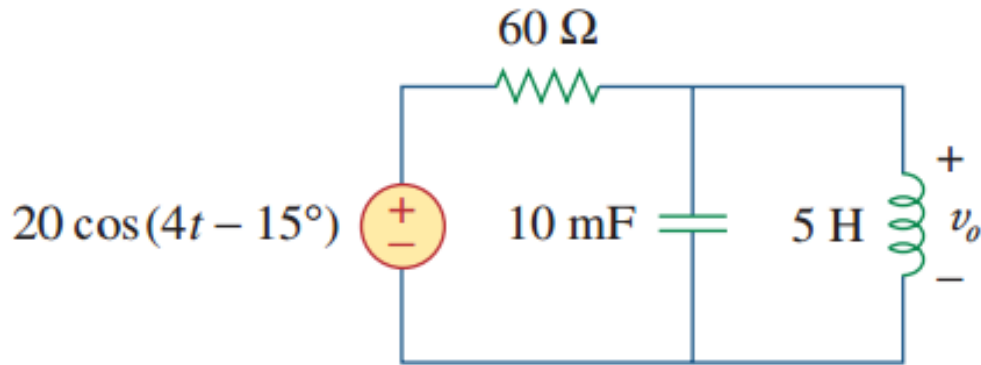
$$Z_3 = 8 + j\omega L = 8 + j50 \times 0.2 = (8 + j10) \Omega$$

$$Z_{\text{in}} = Z_1 + Z_2 \parallel Z_3 = 3.22 - j11.07 \Omega$$

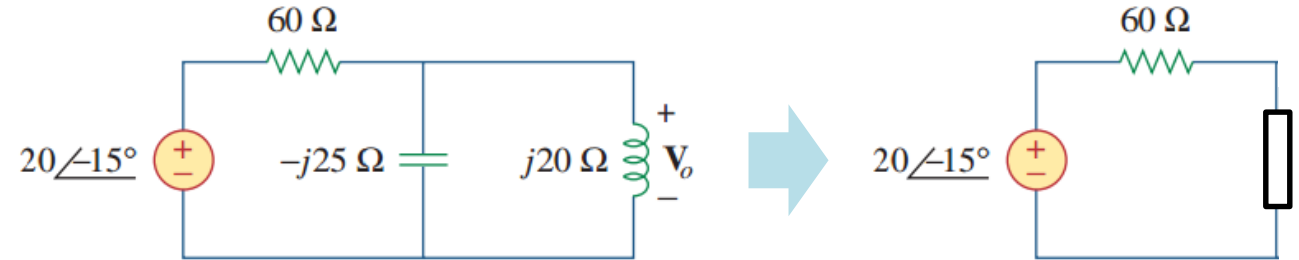
9.5 KVL and KCL in frequency domain (h)

- Example 9.10:

Example 9.10: Determine $v_o(t)$ in the circuit



Solution:



$$\mathbf{Z}_1 = 60 \, \Omega$$

$$\mathbf{Z}_2 = -j25 \parallel j20 = \frac{-j25 \times j20}{-j25 + j20} = j100 \, \Omega$$

$$\begin{aligned} \mathbf{V}_o &= \frac{\mathbf{Z}_2}{\mathbf{Z}_1 + \mathbf{Z}_2} \mathbf{V}_s = \frac{j100}{60 + j100} (20 \angle -15^\circ) \\ &= (0.8575 \angle 30.96^\circ) (20 \angle -15^\circ) = 17.15 \angle 15.96^\circ \, \text{V} \end{aligned}$$

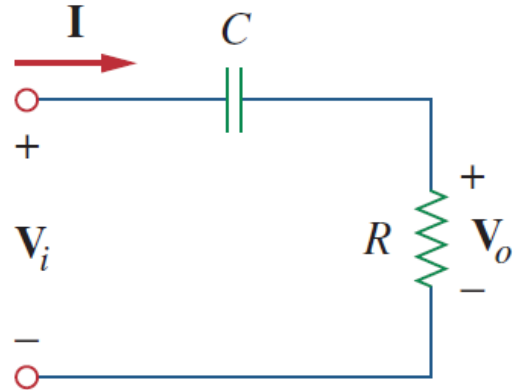
$$v_o(t) = 17.15 \cos(4t + 15.96^\circ) \, \text{V}$$

Lesson 9 Sinusoids and Phasors

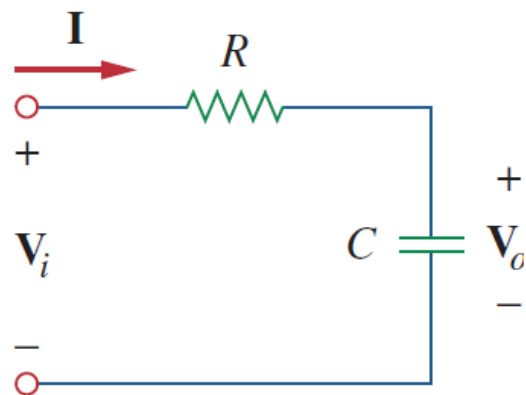
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9.6 Phase shifters (a)

- Phase shifters (移相器):



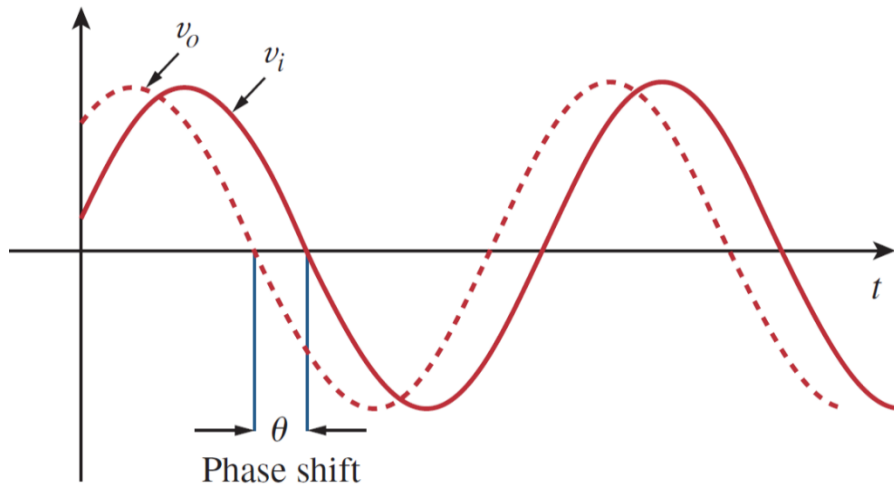
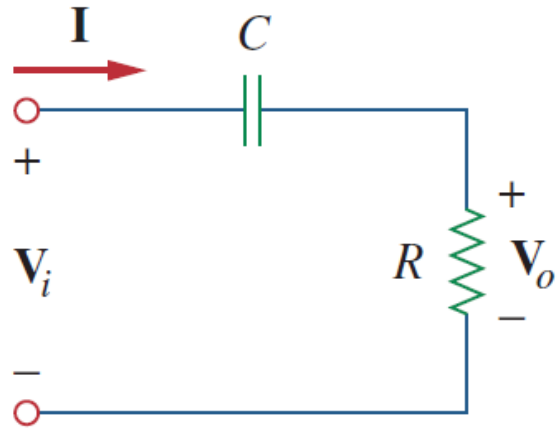
(a)



- A phase-shifting circuit (**phase shifter**) is often employed to correct an undesirable phase shift already present in a circuit or to produce special desired effects.
- An RC circuit is suitable for this purpose because its capacitor causes the circuit current to lead the applied voltage.
- RL circuits or any reactive circuits could also serve the same purpose.

9.6 Phase shifters (b)

- Positive phase shifters, v_o from the resistor:



a) When the output voltage is taken from the resistor: v_o leads v_i

b) If $X_C = -1/\omega C$, then the total impedance is $Z = R + jX_C$,

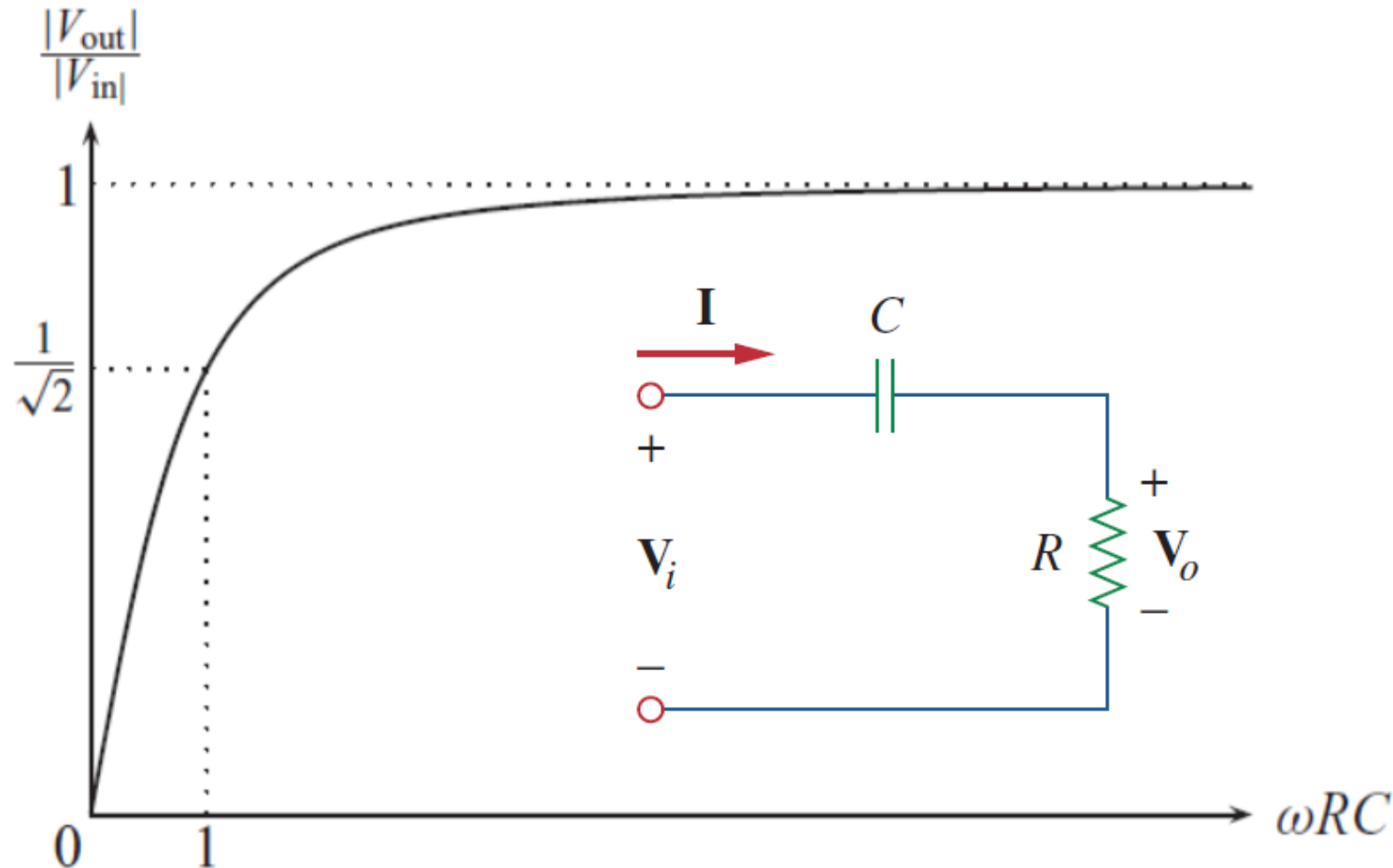
$$\mathbf{V}_o = \frac{R}{R + \frac{1}{j\omega C}} \mathbf{V}_i = \frac{j\omega RC}{j\omega RC + 1} \mathbf{V}_i = \frac{j\omega RC + (\omega RC)^2}{1 + (\omega RC)^2} \mathbf{V}_i$$

$$\theta = \tan^{-1} \left(\frac{1}{\omega RC} \right) = \tan^{-1} \left(\frac{-X_C}{R} \right) \quad |\theta| = 45^\circ \text{ if } R = |X_C|$$

$$\frac{|\mathbf{V}_o|}{|\mathbf{V}_i|} = \frac{\omega RC}{\sqrt{1 + (\omega RC)^2}}$$

9.6 Phase shifters (c)

- Positive phase shifters, v_o from the resistor:



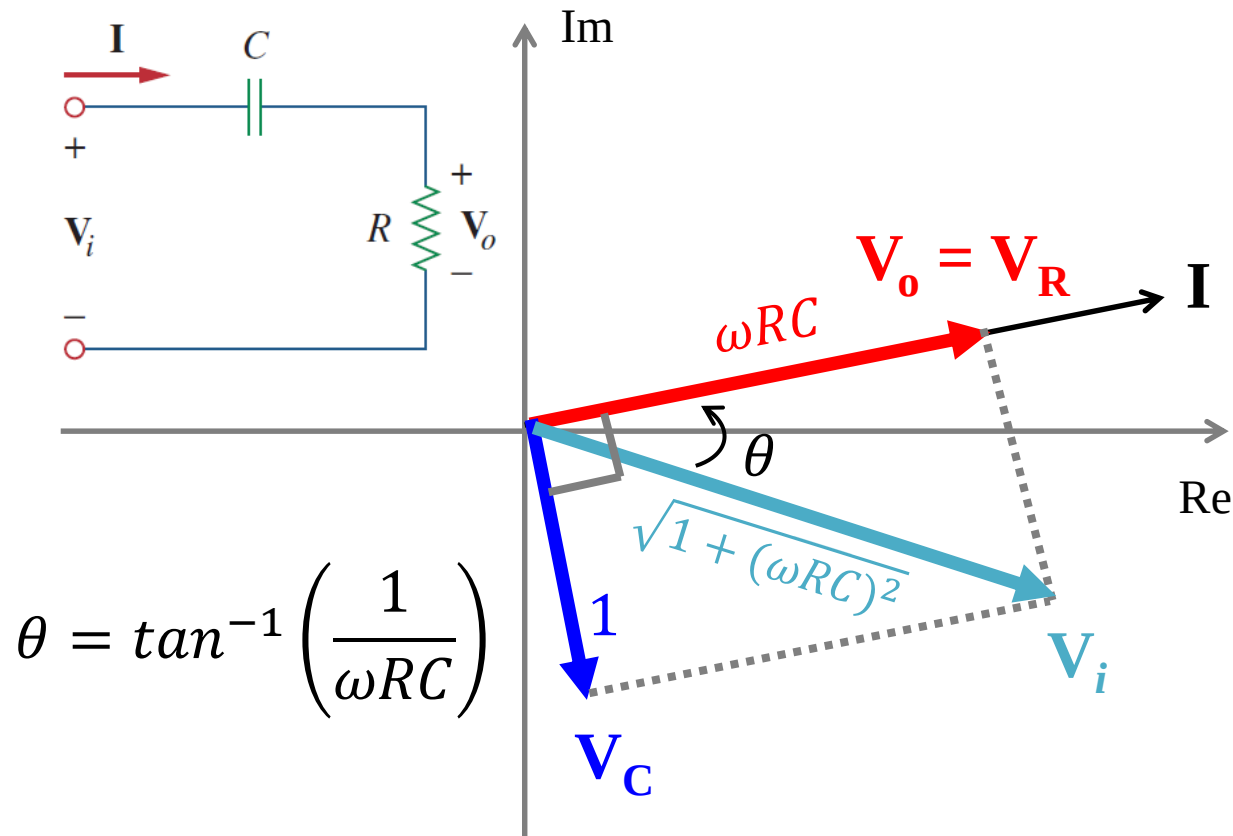
$$\frac{|V_o|}{|V_i|} = \frac{\omega RC}{\sqrt{1 + (\omega RC)^2}}$$

$$\theta = \tan^{-1} \left(\frac{1}{\omega RC} \right)$$

$$\theta \rightarrow \frac{\pi}{2}, \quad \frac{|V_o|}{|V_i|} \rightarrow 0$$

9.6 Phase shifters (d)

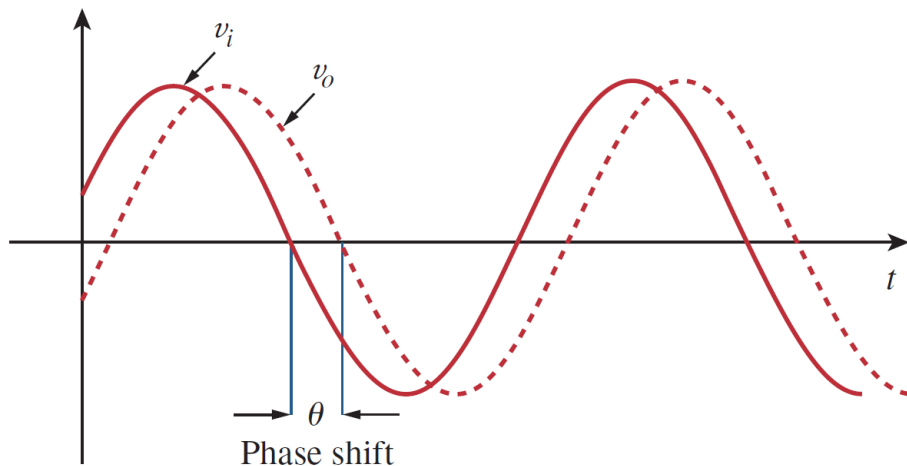
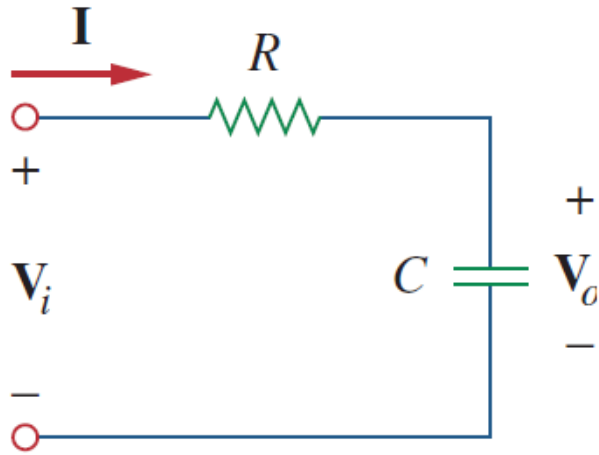
- Positive phase shifters, v_o from the resistor:



- The current I can be used as the phase reference for elements in series, since in such circuit the elements share the same I .
- We don't know the phase of the I , yet we can randomly pick one just for qualitative analysis.
- KVL: $V_i = V_C + V_R$
- The resistor voltage V_R is in phase with the I , as it has no reactance.
- The V_C is 90° behind the I ($I = j\omega V$).
- According to the voltage division, the phase shift is $\tan^{-1}(1/\omega RC)$

9.6 Phase shifters (e)

- Negative phase shifters, v_o from the capacitor:



- a) The output is taken across the capacitor, so v_o lags v_i

$$\mathbf{V}_o = \frac{\frac{1}{j\omega C}}{R + \frac{1}{j\omega C}} \mathbf{V}_i = \frac{1}{j\omega RC + 1} \mathbf{V}_i = \frac{1 - j\omega RC}{1 + (\omega RC)^2} \mathbf{V}_i$$

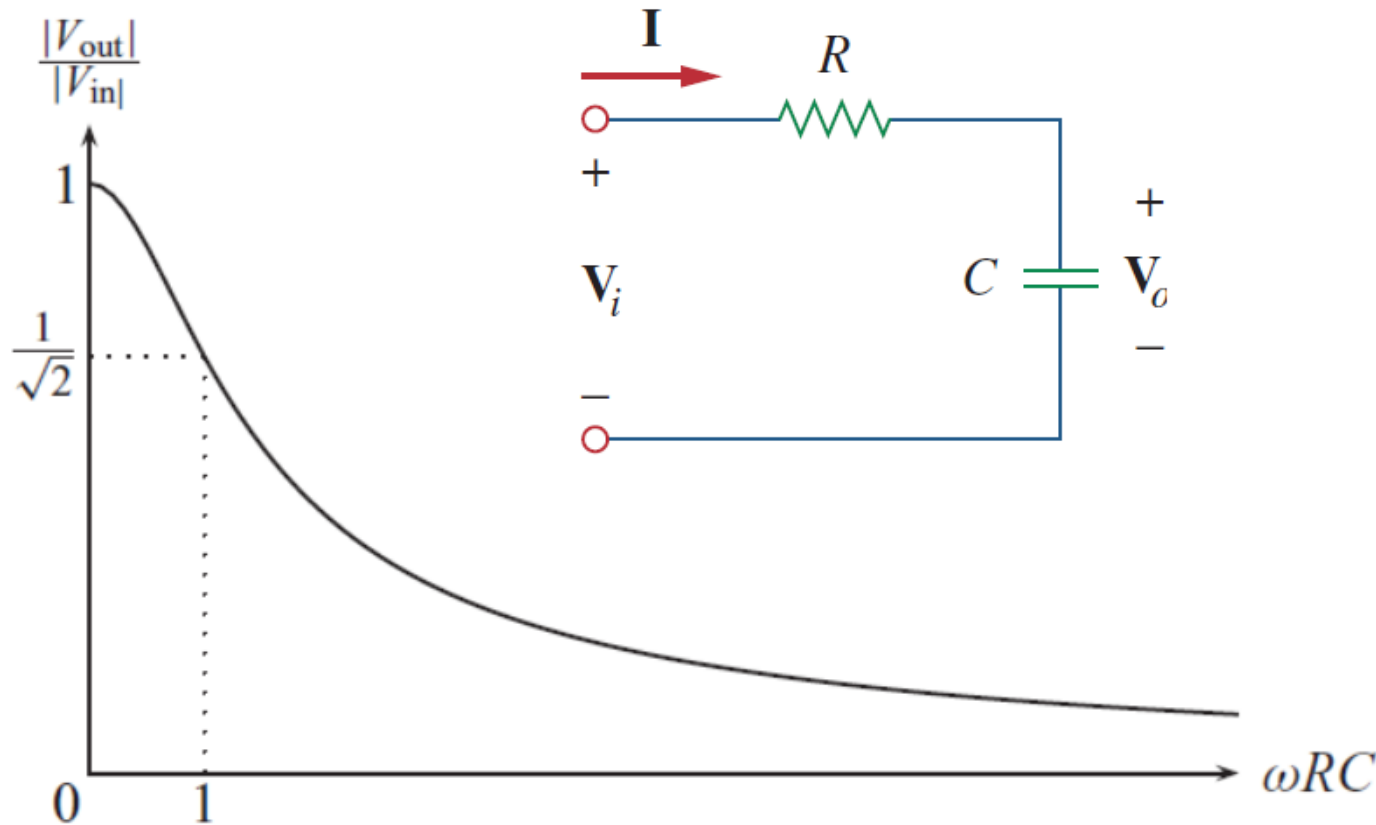
$$\theta = \tan^{-1}(-\omega RC) = \frac{\pi}{2} - \tan^{-1}\left(\frac{X_C}{R}\right)$$

$$|\theta| = 45^\circ \text{ if } R = |X_C|$$

$$\frac{|\mathbf{V}_o|}{|\mathbf{V}_i|} = \frac{1}{\sqrt{1 + (\omega RC)^2}}$$

9.6 Phase shifters (f)

- Negative phase shifters, v_o from the capacitor:



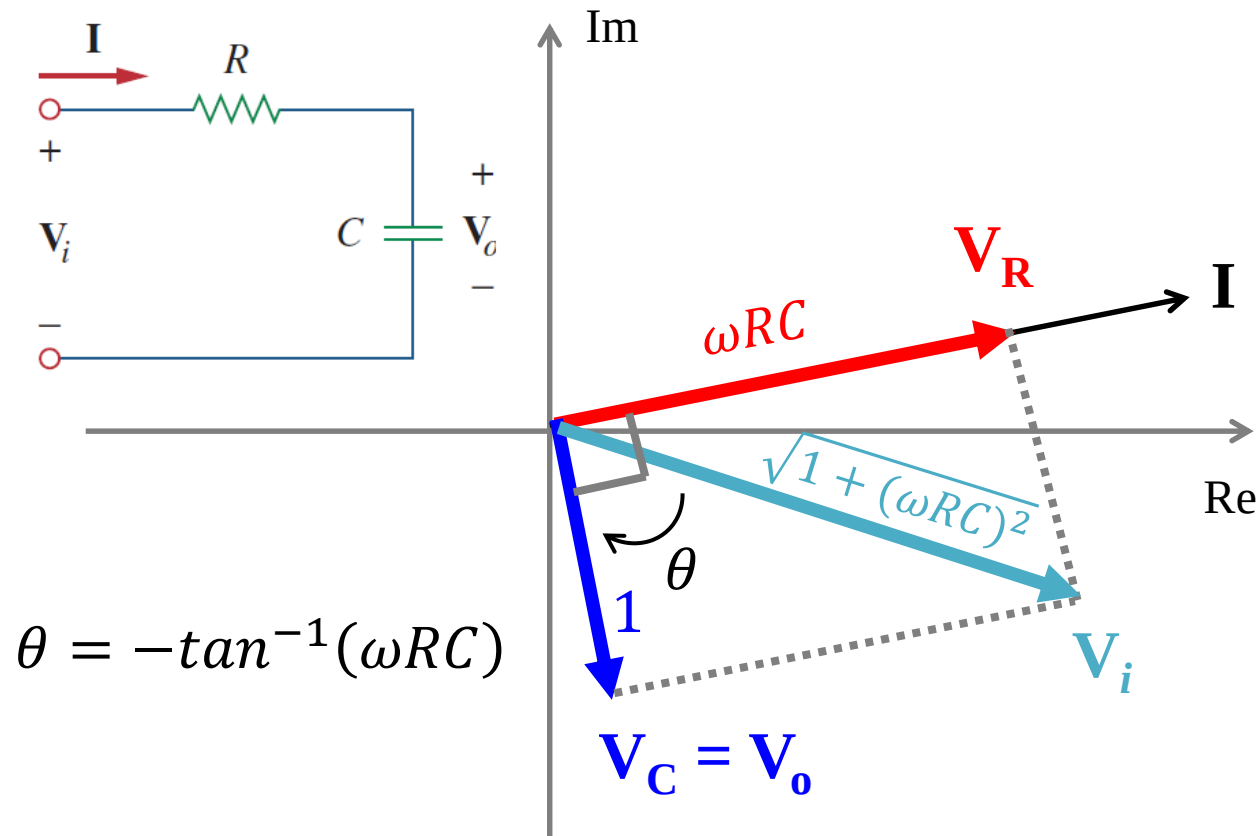
$$\frac{|V_o|}{|V_i|} = \frac{1}{\sqrt{1 + (\omega RC)^2}}$$

$$\theta = \tan^{-1}(-\omega RC)$$

$$\theta \rightarrow \frac{\pi}{2}, \quad \frac{|V_o|}{|V_i|} \rightarrow 0$$

9.6 Phase shifters (g)

- Negative phase shifters, v_o from the capacitor:

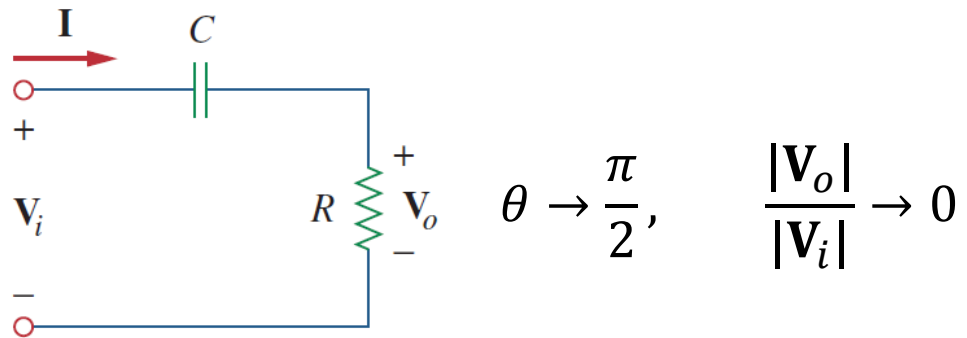


- The current I can be used as the phase reference for elements in series, since in such circuit the elements share the same I .
- We don't know the phase of the I , yet we can randomly pick one just for qualitative analysis.
- KVL: $V_i = V_c + V_R$
- The resistor voltage V_R is in phase with the I , as it has no reactance.
- The V_C is 90° behind the I ($I = j\omega V$).
- According to the voltage division, the phase shift is $-\tan^{-1}(\omega RC)$

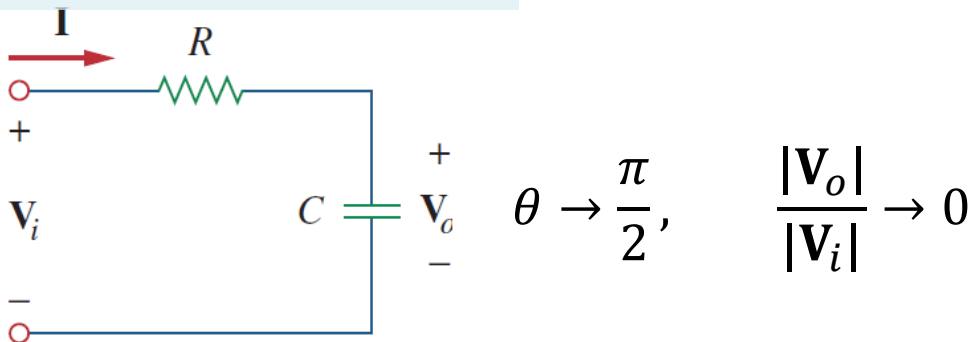
9.6 Phase shifters (h)

- Phase shift versus voltage division:

Positive phase shifter



Negative phase shifter

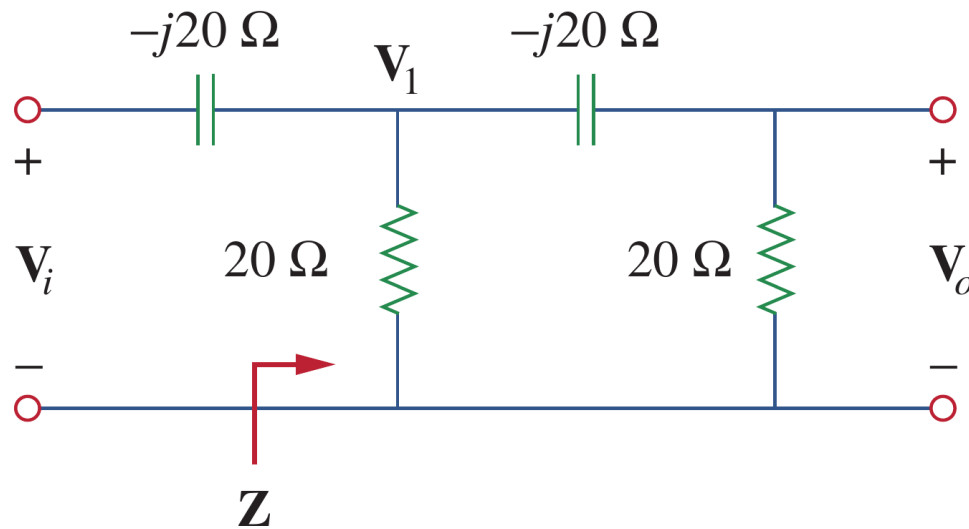


- The simple RC phase shifters also act as voltage dividers, so the V_o approaches zero when the phase shift θ approaches 90° .
- Simple RC circuits are used only when small amounts of phase shift are required. If it is desired to have phase shifts greater than 60° , simple RC networks are cascaded, thereby providing a total phase shift equal to the sum of the individual phase shifts.
- In practice, the phase shifts due to the stages are not equal, because the succeeding stages load down the earlier stages unless op amps are used to separate the stages.

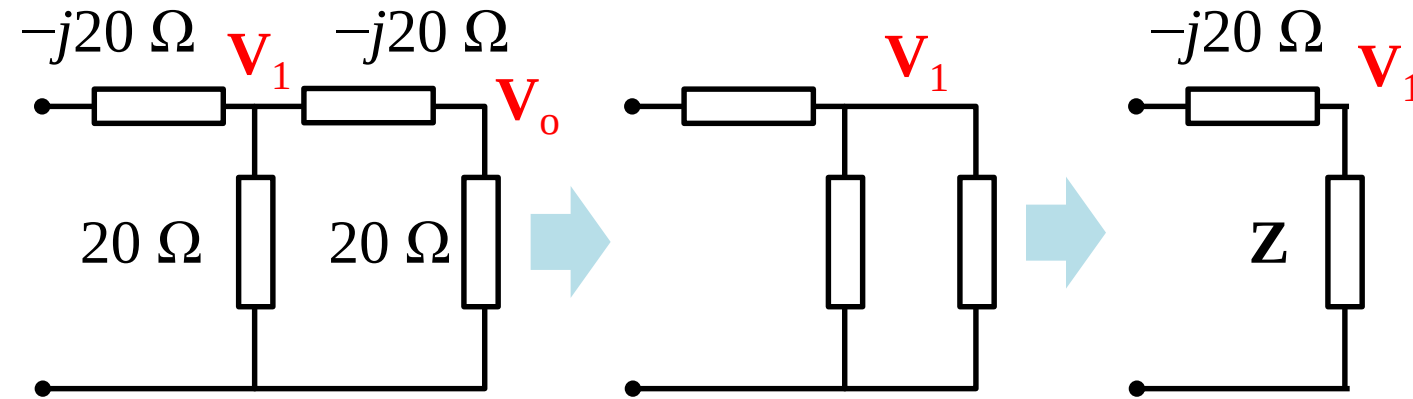
9.6 Phase shifters (i)

- Example 9.11:

Example 9.11: find the phase shift between V_i and V_o in the following circuit.



Solution:



$$Z = 20 \parallel (20 - j20) = \frac{20(20 - j20)}{40 - j20}$$

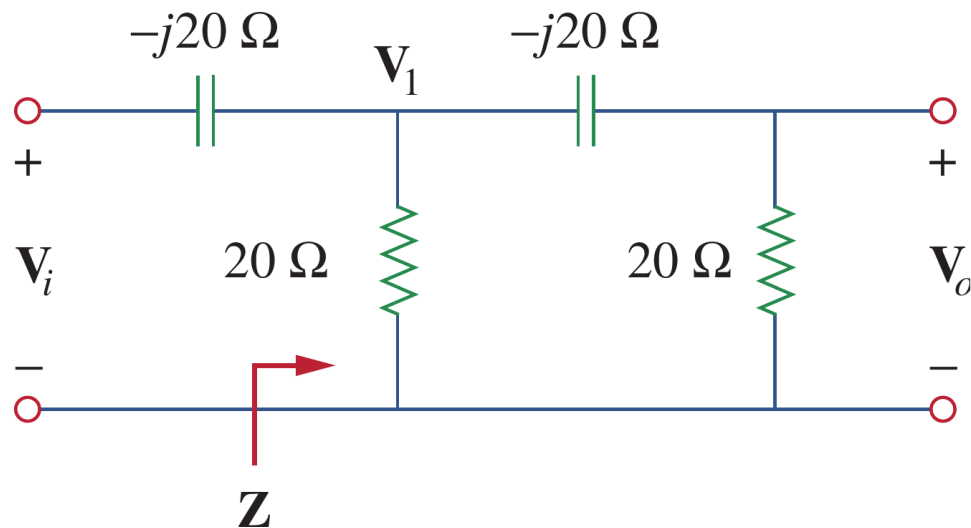
$$= \frac{20(20 - j20)}{40 - j20} \times \frac{40 + j20}{40 + j20} = 12 - 8j \Omega$$

$$V_1 = \frac{Z}{Z - j20} V_i = \frac{12 - j4}{12 - j24} V_i = \frac{\sqrt{2}}{3} \angle 45^\circ V_i$$

9.6 Phase shifters (j)

- Example 9.11:

Example 9.11: find the phase shift between V_i and V_o in the following circuit.



Solution:

$$V_o = \frac{20}{20 - j20} V_1 = \frac{\sqrt{2}}{2} \angle 45^\circ V_1$$

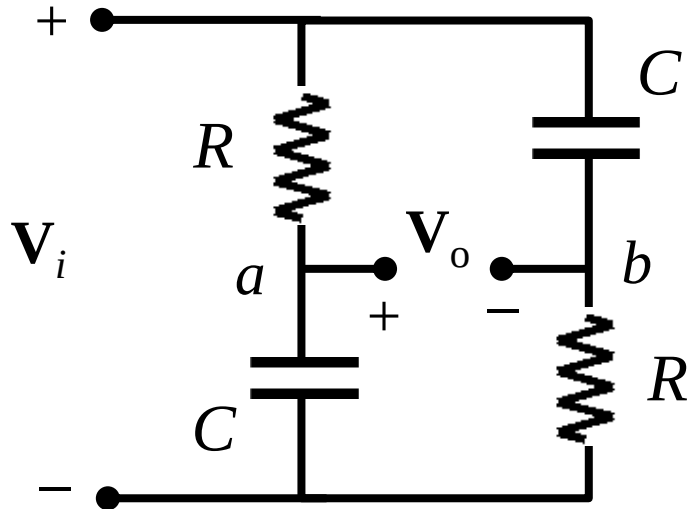
$$V_o = \left(\frac{\sqrt{2}}{2} \angle 45^\circ \right) \left(\frac{\sqrt{2}}{3} \angle 45^\circ V_i \right) = \frac{1}{3} \angle 90^\circ V_i$$

The output of a passive voltage divider or phase shifter will be affected by the load.

9.6 Phase shifters (k)

- Example 9.12:

Example 9.12: find the V_o in the following circuit.



Solution:

According to the voltage division:

$$V_a = \frac{1}{1 + j\omega RC} V_i \quad V_b = \frac{j\omega RC}{1 + j\omega RC} V_i$$

$$V_o = V_a - V_b = \frac{1 - j\omega RC}{1 + j\omega RC} V_i$$

$$|V_o| = \frac{\sqrt{1 + (\omega RC)^2}}{\sqrt{1 + (\omega RC)^2}} |V_i| = |V_i|$$

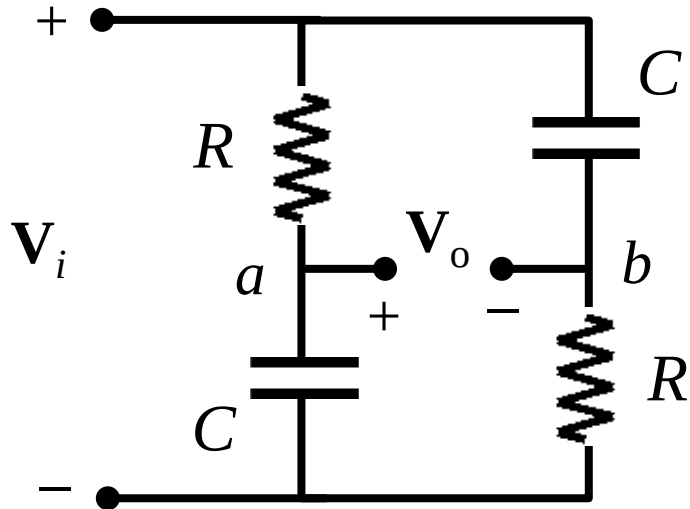
$$\theta = -2\tan^{-1}(\omega RC)$$

Amplitude will not change but the phase can be lagged.

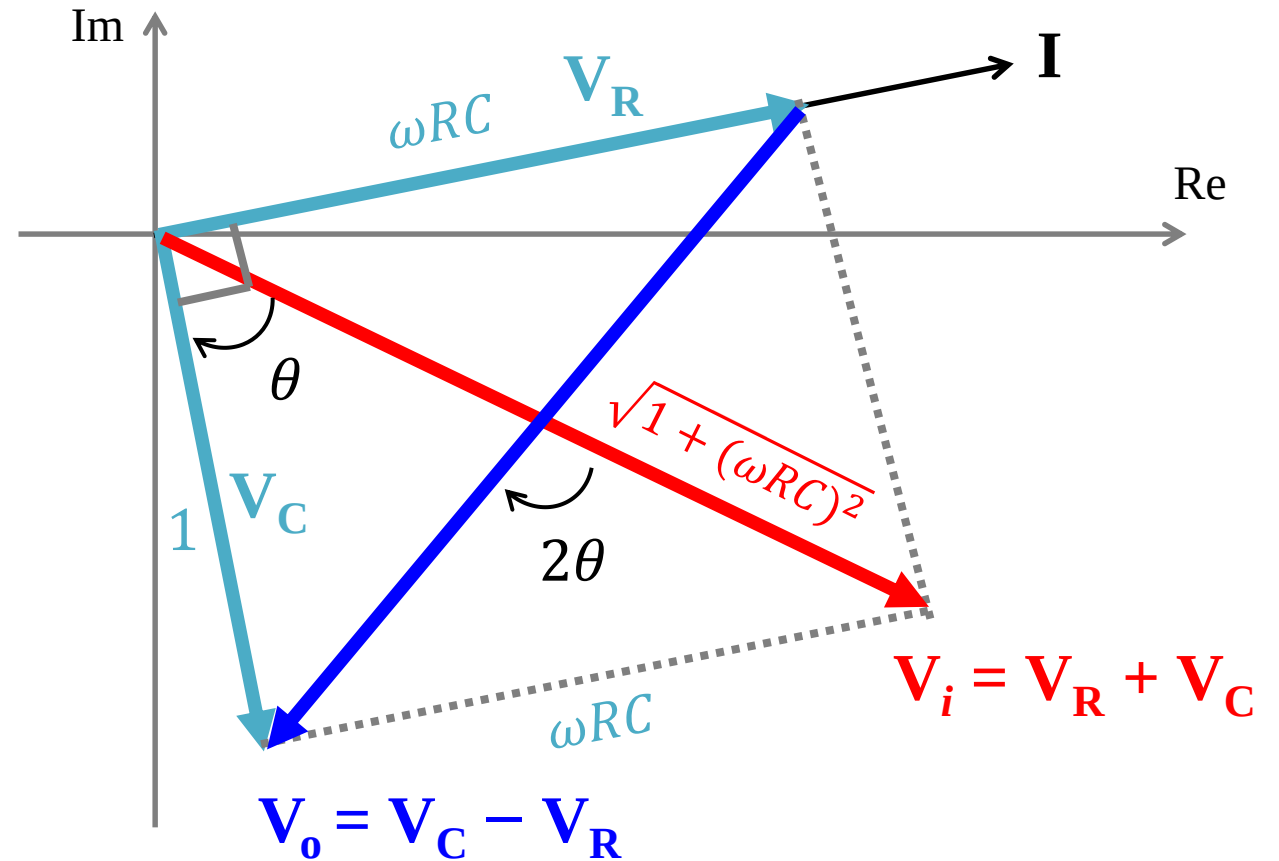
9.6 Phase shifters (I)

- Example 9.12:

Example 9.12: find the V_o in the following circuit.



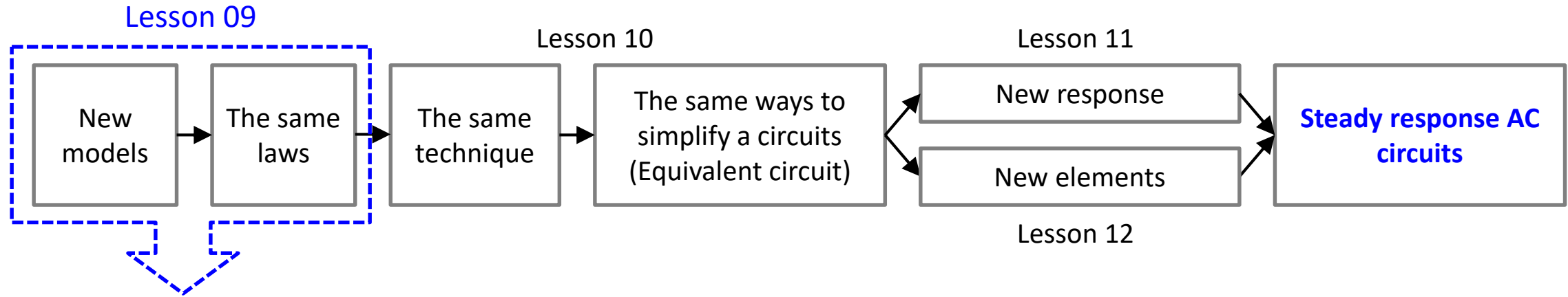
Solution:



Lesson 9 Sinusoids and Phasors

9.1	Introduction	Slides 05-10
9.2-9.3	Sinusoids and phasors	Slides 11-41
9.4	Impedances and admittances in AC	Slides 43-54
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9.6	Phase shifters (AC voltage dividers)	Slides 65-76
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9.7 Summary



1. A sinusoid is a signal in the form of the sine or cosine function, i.e., $v(t) = V_m \cos(\omega t + \phi)$
2. A phasor is a complex quantity that represents both the amplitude and the phase of a sinusoid (missing out the angular frequency), denoted with boldface letter. $\mathbf{V} = V_m \angle \phi$.
3. In an ac circuit, the phase of voltage relative to the current is always fixed: in phase for resistor; leading ahead 90° for inductor and lagging behind 90° for capacitor.
4. Generalized Ohm's law holds for all three type of passive elements.
5. Kirchhoff's laws are still valid for phasors
6. The techniques of voltage/current division, series/parallel combination of impedance/admittance, circuit reduction, and wye-delta transform all apply to ac circuit analysis.

hole-punch cloud

